

ABSTRACT

This thesis presents a multi-agent decision support system for stock trading on the Warsaw Stock Exchange, an underrepresented Central and Eastern European market, and evaluates it under a protocol designed to address four weaknesses that recur in empirical studies of trading strategies: in-sample parameter selection, absent transaction costs, neglect of risk-adjusted performance, and lack of significance tests.

The system consists of five heterogeneous technical-analysis agents, each confined to a single methodological family, feeding a transparent, confidence-weighted decision agent whose aggregation rule is grounded in forecast-combination theory. The evaluation combines anchored walk-forward tuning, calibrated commissions and slippage, the Sharpe, Sortino, Calmar and Martin ratios together with the Ulcer Index, a volatility-matched comparison, bootstrap confidence intervals and paired significance tests, a decomposition of the test window into market regimes, a leave-one-agent-out ablation, sensitivity analyses over the decision-agent weights, the selection objective and transaction costs, and an empirical audit of the confidence heuristic. Two machine-learning baselines, an LSTM directional classifier and a gradient-boosted-trees classifier, are evaluated under the identical protocol.

On 435 held-out trading days for WIG20, the system trails passive buy-and-hold on gross total return in a bull-dominated window, but improves every risk-adjusted metric, halves the maximum drawdown, reduces the integrated drawdown by nearly two-thirds, and overtakes the benchmark on total return once both series are scaled ex ante to the same volatility. These differences are economically meaningful yet not statistically significant on a single sub-two-year window.

Regime analysis attributes the profile to smaller losses during corrections and gains during sideways phases, offset by lower bull-market exposure. The same qualitative pattern appears in an external validation on the FTSE 100 under UK-specific costs, and the system remains competitive with both machine-learning baselines. A dedicated experiment shows that the confidence values carry no measurable information about signal-level accuracy, so the operative mechanism is the pooling of weakly and negatively correlated signals rather than precision weighting. The results support theory-motivated ensembles as risk-aware decision support rather than as return enhancers.

Keywords:

algorithmic trading, decision support systems, multi-agent systems, forecast combination, risk-adjusted performance, walk-forward validation, cross-market validation, Warsaw Stock Exchange.

Field of science and technology in accordance with OECD requirements:

engineering and technology; electrical, electronic and information engineering; computer and information sciences.

STRESZCZENIE

W pracy przedstawiono wieloagentowy system wspomagania decyzji dla handlu akcjami na Giełdzie Papierów Wartościowych w Warszawie, czyli na rynku Europy Środkowo-Wschodniej rzadko analizowanym w literaturze, oraz poddano go ocenie według protokołu usuwającego cztery słabości powtarzające się w badaniach empirycznych strategii inwestycyjnych: dobór parametrów na danych, na których strategia jest oceniana, pominięcie kosztów transakcyjnych, pominięcie miar skorygowanych o ryzyko oraz brak testów istotności statystycznej.

System składa się z pięciu heterogenicznych agentów analizy technicznej, z których każdy wykorzystuje wyłącznie jedną rodzinę metodologiczną wskaźników, oraz z przejrzystego agenta decyzyjnego, który agreguje ich sygnały z wagami wynikającymi z pewności sygnału. Regułę agregacji osadzono w teorii kombinacji prognoz (ang. *forecast combination*). Ocena łączy strojenie parametrów metodą walidacji kroczącej z zakotwiczonym oknem uczącym (ang. *anchored walk-forward*), skalibrowane koszty transakcyjne, miary skorygowane o ryzyko wraz z indeksem Ulcera, porównanie przy zrównanej zmienności, przedziały ufności i sparowane testy istotności metodą bootstrap, dekompozycję okna testowego na reżimy rynkowe, analizę ablacyjną, analizy wrażliwości oraz empiryczną weryfikację heurystyki pewności sygnału. Dwa modele uczenia maszynowego, klasyfikator kierunkowy LSTM oraz klasyfikator oparty na wzmacnianych gradientowo drzewach decyzyjnych, oceniono według identycznego protokołu.

W okresie 435 sesji giełdowych wyłączonych z procesu uczenia system uzyskał dla indeksu WIG20 niższą stopę zwrotu brutto niż pasywna strategia kup i trzymaj w oknie zdominowanym przez trend wzrostowy, poprawił jednak każdą z rozważanych miar skorygowanych o ryzyko, zmniejszył o połowę maksymalne obsunięcie kapitału, ograniczył skumulowane obsunięcie o niemal dwie trzecie oraz przewyższył strategię odniesienia pod względem stopy zwrotu po sprowadzeniu obu szeregów do tej samej zmienności wyznaczonej z góry. Różnice te są istotne ekonomicznie, lecz nie osiągają istotności statystycznej w oknie krótszym niż dwa lata.

Analiza reżimów rynkowych wskazuje, że profil ten wynika z mniejszych strat w okresach korekty i zysków w okresach konsolidacji, okupionych mniejszym udziałem w fazach wzrostowych. Ten sam wzorec potwierdziła walidacja zewnętrzna na indeksie FTSE 100, a system pozostał konkurencyjny wobec obu modeli uczenia maszynowego. Odrębny eksperyment wykazał, że wartości pewności sygnału nie niosą mierzalnej informacji o trafności pojedynczych sygnałów, zatem mechanizmem działania systemu jest łączenie sygnałów słabo i ujemnie skorelowanych, a nie wążenie precyzją. Wyniki potwierdzają przydatność zespołów agentów jako narzędzia wspomagania decyzji świadomego ryzyka, a nie metody zwiększania stopy zwrotu.

Słowa kluczowe:

handel algorytmiczny, systemy wspomagania decyzji, systemy wieloagentowe, kombinacja prognoz, efektywność skorygowana o ryzyko, walidacja krocząca, walidacja międzyrynkowa, Giełda Papierów Wartościowych w Warszawie.

Dziedzina nauki i techniki, zgodnie z wymogami OECD:

nauki inżynieryjne i techniczne; elektrotechnika, elektronika, inżynieria informatyczna; nauki o komputerach i informatyka.

LIST OF IMPORTANT SYMBOLS AND ABBREVIATIONS

Symbols of the forecast-combination model

f_k	$[\cdot]$	–	forecast produced by the k -th forecaster
e_k	$[\cdot]$	–	forecast error of the k -th forecaster, $e_k = y - f_k$
σ_k^2	$[\cdot]$	–	variance of the forecast error of the k -th forecaster
$\rho_{k,j}$	$[\cdot]$	–	correlation between the errors of forecasters k and j
w_k	$[\cdot]$	–	weight assigned to the k -th forecaster in the combination
K	$[\cdot]$	–	number of forecasters in the pool
y	$[\cdot]$	–	unobserved target of the forecast

Symbols of the decision layer

$\sigma_{a,s}(t)$	$[\cdot]$	–	directional signal of agent a for symbol s at date t , $\in \{-1, 0, +1\}$
$c_{a,s}(t)$	$[\cdot]$	–	confidence of agent a for symbol s at date t , $\in [0, 1]$
$\kappa(a)$	$[\cdot]$	–	methodological family of agent a
w_κ	$[\cdot]$	–	family-level weight of family κ
$c_{\min}$	$[\cdot]$	–	minimum confidence below which a signal is discarded
$\theta_{\text{buy}}, \theta_{\text{sell}}$	$[\cdot]$	–	decision thresholds of the decision agent
m	$[\cdot]$	–	decision-agent mode: passive, balanced or aggressive

Symbols of the indicators

h	$[\cdot]$	–	value of the MACD histogram
σ	$[\cdot]$	–	standard deviation over a rolling window
τ	$[\cdot]$	–	threshold of the rate-of-change rule
U, L	$[\cdot]$	–	upper and lower Bollinger band
m	$[\cdot]$	–	Bollinger band multiplier
RS	$[\cdot]$	–	relative strength, the ratio of average gain to average loss

Symbols of the evaluation

r_t	$[\cdot]$	– portfolio return on day t
$\bar{r}$	$[\cdot]$	– mean daily return
N	$[\cdot]$	– number of observations in the evaluation window
dd_t	$[\cdot]$	– running drawdown of the equity curve at date t
λ	$[\cdot]$	– ex-ante leverage factor of the volatility-matched comparison
σ^{target}	$[\cdot]$	– target annualised volatility of the benchmark
σ^{strat}	$[\cdot]$	– annualised volatility of the strategy on the training window
γ	[bps]	– proportional commission rate
s	[bps]	– one-way slippage rate
$T_{\text{tr}}, T_{\text{te}}$	$[\cdot]$	– training and test windows
ρ	$[\cdot]$	– Spearman rank correlation coefficient
δ	$[\cdot]$	– half-width of the no-trade band of a machine-learning baseline
h	[d]	– prediction horizon of a machine-learning baseline
n	$[\cdot]$	– number of pooled signal observations

Abbreviations used in the thesis

B&H	– buy-and-hold, the passive benchmark strategy
bps	– basis point, one hundredth of a percentage point
CAGR	– compound annual growth rate
CEE	– Central and Eastern Europe
CI	– confidence interval
CSV	– comma-separated values, the price-data file format
DSS	– decision support system
EMA	– exponential moving average
FTSE 100	– Financial Times Stock Exchange 100 Index (London Stock Exchange)
GBM	– gradient boosting machine, the gradient-boosted-trees baseline
LSTM	– long short-term memory, a recurrent neural-network architecture
MACD	– moving-average convergence-divergence
MAS	– multi-agent system
MDD	– maximum drawdown
OHLCV	– open, high, low, close and volume price data
ROC	– rate of change
RSI	– relative strength index
SDRT	– Stamp Duty Reserve Tax, the UK buy-side transaction tax
SLR	– systematic literature review
SMA	– simple moving average
TR	– total return
UI	– Ulcer Index
UML	– unified modeling language
WIG20	– Warszawski Indeks Gieldowy 20, the WSE blue-chip index
WSE	– Warsaw Stock Exchange
YAML	– YAML Ain't Markup Language, the configuration file format

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1. INTRODUCTION

Decision support systems and algorithmic trading occupy a central place in contemporary financial markets. They help both institutional and retail investors process large volumes of market information and translate that information into trading actions [1]. Over the last two decades, research on algorithmic trading has expanded from simple rule-based strategies [2] to machine-learning classifiers [3, 4], and more recently to deep reinforcement learning systems that learn trading policies directly from data [5, 6, 7].

In parallel, multi-agent systems (MAS) have been proposed as a natural way to organise this diversity of trading rules and models, so that independent agents can produce signals that a coordinator combines into the final decision [8, 9, 10, 11]. The appeal of the arrangement is that it separates two concerns that are easily confused: what an individual indicator says about the market, and what a portfolio should do about it.

The practical problem that motivates this thesis is narrower than the breadth of that literature suggests. A retail investor on the Warsaw Stock Exchange (WSE) has access to daily price data, a modest brokerage account, and a large menu of published trading rules, but has no reliable way to tell which of those rules, or which combination of them, is worth following. The published evidence is not much help, because a substantial part of it is produced under evaluation protocols that make results look better than they are.

1.1. Problem statement

Four methodological weaknesses recur in empirical studies of trading strategies, and together they undermine the credibility of reported results.

The first is *in-sample parameter selection*. When the best configuration of a strategy is chosen on the same data used to evaluate it, the out-of-sample performance is overstated [12, 13]. The second is the *omission of realistic frictions*, in particular commissions and slippage, which can erode a noticeable fraction of the gross return of high-turnover configurations.

The third is the *neglect of risk-adjusted performance*. Many studies compare a trading strategy to passive buy-and-hold purely on cumulative profit, without taking into consideration that an active strategy can produce a smoother equity curve at a lower unleveraged return. This weakness is particularly acute during sustained bull markets, where passive exposure mechanically dominates any strategy that is occasionally out of the market [14]. The fourth is the *absence of formal significance tests*. Headline metrics are often reported as point estimates without confidence intervals or paired tests against the benchmark, even though trading-strategy metrics are known to be non-Gaussian and path-dependent [15, 12].

Beyond these methodological issues, the empirical evidence on multi-agent decision support systems is strongly concentrated on major markets in the United States, Western Europe, and Asia. The Warsaw Stock Exchange, the largest exchange in Central and Eastern Europe (CEE) by capitalisation, has received much less attention [16, 17, 18]. A retail investor on that market therefore faces a double difficulty: the available evidence is methodologically fragile, and what evidence exists was gathered elsewhere.

1.2. Aim and scope of the thesis

The aim of this thesis is to design a transparent multi-agent decision support system for trading large-capitalisation stocks on the Warsaw Stock Exchange, and to determine, under an evaluation protocol that addresses the four weaknesses stated above, whether such a system offers a measurable advantage over passive investment.

The scope is defined by four decisions. The system trades daily closing prices of WIG20 index constituents, so intraday behaviour, order-book dynamics, and other asset classes are outside its remit. The signal agents use only price-derived technical indicators, so fundamental and alternative data are excluded. The aggregation rule is a linear, confidence-weighted score, chosen so that the decision layer remains interpretable and its mapping to forecast-combination theory stays transparent, in contrast to opaque end-to-end learners. Finally, all evidence is produced by simulation on historical data, so no live-trading exercise is part of the work.

Three hypotheses, stated formally in Section 3.3 and tested in Chapter 5, give the aim its testable form. In summary, they assert that the system matches or exceeds passive buy-and-hold on risk-adjusted metrics even where it does not on gross return (H1), that it outperforms the best single-agent baseline on total return (H2), and that its performance is robust to the removal of any single agent (H3).

The technical assumptions adopted throughout are the following. Positions are opened and closed at the daily closing price. Transaction costs comprise a proportional commission and one-way slippage, calibrated per market. Only decision-agent parameters are tuned, and only on a training window that precedes the evaluation window chronologically; all indicator hyperparameters are fixed at their literature-conventional values. Portfolio capital is allocated as a fixed fraction of available cash per position, without leverage.

1.3. Own contribution

The work reported in this thesis is the author's own. The system, the experiments, and the analysis were designed, implemented, and carried out by the author. Specifically, the author's contribution comprises the following:

- the design and implementation of the complete software system described in Chapter 4, comprising five signal agents, the decision agent, the portfolio simulator, the metrics calculator, and the experiment harness,
- the design of the evaluation protocol of Chapter 5, including the anchored walk-forward procedure, the transaction-cost calibration, the market-regime labelling, the volatility-matched comparison, and the bootstrap testing scheme,
- the implementation and in-sample tuning of the two machine-learning baselines used as reference points,
- the conceptual grounding of the aggregation rule in forecast-combination theory, together with the correlation and confidence-validation experiments that test how far that grounding holds empirically,
- the execution and interpretation of all experiments reported in Chapter 5, including the external validation on the FTSE 100.

An earlier version of parts of this material was prepared by the author for publication in a co-authored journal article; the text of this thesis was written by the author for this purpose.

1.4. *Structure of the thesis*

The thesis is organised into six chapters, whose significance is unequal by design: Chapters 4 and 5 carry the original work, while the remaining chapters frame it.

Chapter 2 reviews the state of the art in four streams: algorithmic trading and technical analysis, multi-agent systems in finance, algorithmic trading on Central and Eastern European markets, and risk-adjusted performance evaluation. It establishes both the methodological gap and the empirical niche that the thesis addresses.

Chapter 3 presents the theoretical background and the system design. It introduces forecast-combination theory and the variance-reduction argument that motivates the architecture, maps that theory onto the trading problem, states the three hypotheses, and then describes the four-layer architecture, the five signal agents, the decision agent, and the agent taxonomy.

Chapter 4 describes the software realisation: the requirements it had to satisfy, the technologies selected and why, the structure of the code, the data preparation pipeline, and the way the implementation was verified.

Chapter 5 reports the empirical part of the work. It describes the evaluation protocol in full, reports the results on the WIG20 panel, subjects them to a battery of robustness checks and an external validation on the FTSE 100, and interprets the evidence against the three hypotheses.

Chapter 6 summarises what was achieved, states the conclusions as an organised list, discusses the limitations of the study and the difficulties encountered during the work, and indicates directions for further development.

2. STATE OF THE ART

This chapter reviews four streams of literature that inform the work reported in this thesis: algorithmic trading and technical analysis, multi-agent systems in finance, algorithmic trading on Central and Eastern European markets, and risk-adjusted performance evaluation. The division into four streams clarifies where the thesis sits in a crowded research landscape, and why the combination of these four angles is the gap it addresses.

2.1. Review method

[Placeholder: this section will be replaced by the protocol and results of the systematic literature review.] The review presented in the remainder of this chapter covers work published between 1966 and 2026, combining the classical contributions that established each stream with recent publications that document its current state.

2.2. Algorithmic trading and technical analysis

Technical analysis uses indicators computed from price and volume to generate trading signals. Early work established that simple trading rules based on moving averages and trading ranges exhibit predictive power that is hard to reconcile with the strict form of the efficient markets hypothesis [2]. Subsequent studies formalised this debate with algorithmic backtests and statistical tests [19], and showed that several popular indicators, among them MACD, RSI and Bollinger Bands, continue to produce information when combined with machine-learning classifiers [3].

Closely related to the ensemble design studied here, empirical work on cross-sectional momentum [20, 21] and time-series momentum [22] has shown that trend-following and mean-reversion signals carry orthogonal information about future returns. That orthogonality is the empirical foundation on which the agent taxonomy of Section 3.7 rests.

More recent work has shifted to deep learning, in particular to recurrent architectures for price prediction [4], and to deep reinforcement learning systems that learn trading policies end-to-end [5, 6, 7, 23]. Systematic reviews conclude that the evidence on technical analysis is mixed, and that it depends strongly on the choice of market, period, and evaluation protocol [24, 25].

To situate the transparent forecast-combination design against this learning-based stream, the evaluation includes two learning-based baselines run under the same walk-forward protocol, transaction costs, and portfolio simulator as the ensemble: an LSTM directional classifier in the style of [4], and a gradient-boosted-trees classifier [26, 27]. The latter belongs to a model family that large-scale comparisons identify as among the strongest learners on tabular financial features [28, 29].

Two methodological criticisms recur across this literature. The first is backtest overfitting: when a large number of configurations are evaluated on the same data, the best configuration is biased upward and the out-of-sample performance is overstated [12, 13]. The second is the frequent absence of realistic frictions, such as commissions and slippage, which can erode a large fraction of the gross return of a high-turnover strategy. Both criticisms are taken seriously here and are built into the evaluation design of Section 5.1.

2.3. Multi-agent systems in finance

Multi-agent systems provide a natural abstraction for trading, because different strategies can be encapsulated as independent agents that communicate through well-defined interfaces. Agent-based artificial markets have been used to study the emergence of liquidity, volatility, and price-discovery patterns from the interaction of heterogeneous traders [30, 31].

Multi-agent decision support systems for stock and currency markets were proposed as early as two decades ago, typically with specialised agents that generate buy and sell recommendations and a coordinating component that aggregates them [8, 9, 10, 11]. The A-Trader system [9, 10] is a particularly close ancestor of the design presented here, since it also uses multiple specialised agents and a consensus-based coordinator, but it has been evaluated primarily on FOREX data. Recent work has extended the paradigm with multi-agent reinforcement learning for portfolio management [5] and algorithmic trading [6].

Outside finance, a substantial body of control-theory research studies how interacting agents reach agreement through consensus protocols, including consensus tracking over cooperation-competition networks [32], convergence-rate analysis under communication delays [33], and logic-based switching schemes for uncertain nonlinear multi-agent dynamics [34]. That literature analyses the dynamical convergence of agent states to a common value over time, whereas the coordinator studied here is a static forecast-combination rule that aggregates heterogeneous directional forecasts at each date. The two notions of agreement are related in spirit but formally distinct.

Relative to this literature, the aggregation mechanism itself is not claimed as novel: combining specialised agents through a weighted score is well established in the systems cited above. The distinction of the present work is twofold. First, it makes the conceptual connection between the aggregator and forecast-combination theory [35, 36, 37] explicit, interpreting the coordinator score as a weighted sum of heterogeneous directional forecasts. That interpretation is used both in the design of the decision agent and in the motivation of the ablation and correlation experiments, and its limits are tested empirically through the confidence-validation and weight-sensitivity analyses of Sections 5.2.8 and 5.3.2.

Second, and more importantly, the resulting ensemble is subjected to a rigorous evaluation protocol, described in Section 5.1, that most prior multi-agent trading studies do not apply in combination. The intended contribution is therefore the theoretical grounding and the evaluation, not the aggregation rule.

2.4. Algorithmic trading on Central and Eastern European markets

Although the Warsaw Stock Exchange is the largest market in Central and Eastern Europe, it is underrepresented in the algorithmic trading literature. Studies of the WSE have focused on the interdependence between the Polish and other stock markets and its macroeconomic drivers [16], on tests of the informational efficiency of the exchange [17], and on the interplay between algorithmic trading, liquidity, and volatility for WSE blue-chip stocks [18].

To the best of the author’s knowledge, multi-agent decision support systems have not been evaluated on WIG20 under an explicit walk-forward protocol with transaction costs and a risk-adjusted comparison. This gap is the primary empirical niche addressed by the thesis. To test whether the resulting profile is specific to WSE conditions, the full pipeline is additionally replicated on the FTSE 100 under UK-specific transaction costs (Section 5.3.7).

2.5. Risk-adjusted performance evaluation

A separate literature has developed a rich set of risk-adjusted metrics that go beyond the Sharpe ratio [38]. The Sortino ratio [39] penalises only downside deviation and is therefore closer to how investors experience losses, and the Calmar ratio [40] relates annualised return to maximum drawdown, which captures the worst realised loss along the equity path. Drawdown-based measures have been extended by the Ulcer Index and the Martin ratio [41], which aggregate the full history of drawdowns rather than only the peak-to-trough extremum.

The adaptive markets hypothesis of [42] argues that market efficiency is itself time-varying, which provides a theoretical motivation for evaluating a strategy across different market regimes rather than only on aggregate metrics. In line with this view, the ensemble is evaluated here both on standard risk-adjusted ratios and across distinct market regimes, and its equity curve is compared against the benchmark after matching the two on ex-ante volatility. This risk-aware evaluation protocol is one of the methodological contributions of the thesis.

2.6. Chapter summary

The four streams reviewed above converge on a single observation. Technical trading rules and multi-agent architectures for combining them are both well established, but the empirical evidence about them is fragile, because it is frequently produced under evaluation protocols that select parameters in sample, ignore transaction costs, report only cumulative profit, and omit significance tests. The risk-adjusted evaluation literature supplies the metrics needed to repair the third of these weaknesses, and the forecast-combination literature supplies a theory that explains what a coordinator of heterogeneous agents should be expected to achieve. What is missing is a study that brings these elements together on a market outside the small set of heavily studied exchanges. That is the gap the remainder of this thesis addresses.

3. THEORETICAL BACKGROUND AND SYSTEM DESIGN

This chapter establishes the theory that motivates the design of the system and then describes the system itself. Sections 3.1 to 3.3 introduce forecast-combination theory, map it onto the trading problem, and state the three hypotheses that the experiments test. Sections 3.4 to 3.7 describe the architecture, the five signal agents, the decision agent that aggregates them, and the taxonomy that makes the heterogeneity of the agent pool explicit.

Throughout the thesis, the term *ensemble* denotes the complete five-agent system together with its decision agent, as opposed to the single-agent and leave-one-agent-out variants used as baselines in Chapter 5.

3.1. Forecast combination and variance reduction

The design of the decision agent is motivated by the theory of forecast combination. Let $f_1, \dots, f_K$ denote K forecasts of an unobserved target y . If each f_k has forecast error $e_k = y - f_k$ with variance σ_k^2 and pairwise correlations ρ_{kj} , then for any convex combination with weights $w_k \geq 0$, $\sum_k w_k = 1$, the variance of the combined error

$$\text{Var}\left(y - \sum_{k=1}^K w_k f_k\right) = \sum_{k=1}^K \sum_{j=1}^K w_k w_j \rho_{kj} \sigma_k \sigma_j \quad (3.1)$$

is generally smaller than the variance of each individual forecast whenever the forecasts are not perfectly correlated [35, 36, 37]. In the special case of uncorrelated forecasts, the optimal weights are proportional to the inverse variances, $w_k \propto 1/\sigma_k^2$. Extensive empirical work has shown that simple combinations, even equal-weight averages, tend to outperform individual forecasts in a wide variety of settings [36, 37].

Two qualitative implications of (3.1) are particularly relevant for the design of a multi-agent trading system. First, negative pairwise correlations reduce the combined variance below what would be achievable with only non-negative correlations, which motivates the explicit inclusion of methodologically complementary agents. A momentum-family agent and a mean-reversion-family agent, for example, are expected to produce negatively correlated signals in trendless markets.

Second, the variance reduction persists as long as at least two imperfectly correlated forecasters remain in the pool. That property motivates the leave-one-agent-out ablation reported in Section 5.3.1.

3.2. Mapping to the trading problem

The trading problem maps onto this framework as follows. At a given date t and symbol s , each signal agent a produces a directional signal $\sigma_{a,s}(t) \in \{-1, 0, +1\}$ together with a confidence value $c_{a,s}(t) \in [0, 1]$. The confidence is read as a heuristic, monotone, bounded proxy for the inverse variance of the agent as a forecaster of the next-period directional move: higher confidence means the indicator of the agent is further inside its active region, which the design treats as a sign that the forecast is more reliable at that moment.

The decision agent aggregates these forecasts into a combined score

$$\text{score}(t, s) = \sum_a w_{\kappa(a)} c_{a,s}(t) \sigma_{a,s}(t), \quad (3.2)$$

where $\kappa(a) \in \{\text{trend, momentum, mean-reversion, volatility}\}$ is the methodological family of agent a , and $w_{\kappa(a)}$ is a family-level weight that plays the role of a prior on the relative reliability of that family. The family weights and the decision-agent thresholds are defined in Section 3.6. The score is then thresholded to produce a buy, sell, or hold decision per symbol.

Two caveats define the status of this mapping and should be kept in mind throughout the thesis. First, the signals and confidence values are heuristic transformations of price history, not statistically calibrated forecasts: no agent estimates a predictive distribution, and no confidence value is fitted to data. The connection to classical forecast-combination theory is therefore *conceptual* rather than literal. Equation (3.1) motivates the architecture, that is, methodologically heterogeneous agents whose errors are imperfectly correlated, and it motivates the linear form of the score (3.2), but it does not by itself guarantee that $c_{a,s}(t)$ behaves as an inverse-variance estimate.

Second, whether higher confidence in fact identifies more accurate signals is an empirical question, which Section 5.2.8 tests directly by relating confidence values to realised directional accuracy on the training window. The hypotheses below are deliberately formulated so that they do not depend on the calibration of the confidence heuristic: they concern the risk-adjusted performance of the pooled ensemble, which (3.1) predicts from error diversity alone.

3.3. Hypotheses

This mapping motivates three testable hypotheses. The first frames the ensemble as a *risk-aware* decision support tool, consistent with the prediction of forecast-combination theory that combining imperfect forecasters reduces variance rather than systematically raising the mean.

- **H1.** On held-out data, net of transaction costs, the multi-agent system matches or exceeds buy-and-hold WIG20 on risk-adjusted metrics (Sharpe, Sortino, Calmar, Ulcer, Martin), on maximum drawdown, and on a volatility-matched comparison of equity curves, even if it does not dominate on gross total return.
- **H2.** On held-out data, net of transaction costs, the multi-agent system outperforms the best single-agent baseline on total return. The rationale is that the combined score diversifies across agents with non-identical errors, so it should dominate any single agent except in the degenerate case where one agent is uniformly best.
- **H3.** The performance of the ensemble is robust to the removal of any single signal agent. The rationale is that no single agent carries the whole directional signal, and forecast-combination theory predicts that variance reduction persists as long as at least two imperfectly correlated forecasters remain.

H1 is stated against buy-and-hold WIG20, because that is the primary panel; Section 5.3.7 replicates the same risk-adjusted comparisons on the FTSE 100 as supplementary external evidence. H2 and H3 are evaluated on WIG20 only. Chapter 5 reports the evidence for each hypothesis and discusses the degree to which it is consistent with the theoretical predictions.

3.4. System architecture

The system is organised into four layers, as shown in Fig. 3.1. The *data layer* loads daily closing prices for the WIG20 constituents into a common panel format and handles missing values by dropping incomplete rows. The *signal layer* hosts the five independent signal agents. Each agent

receives the history of a single symbol up to and including the current date and returns a triple: a directional signal in $\{-1, 0, +1\}$, a confidence value in $[0, 1]$, and a methodological family (trend, momentum, mean-reversion, or volatility).

The *decision layer* is a single decision agent that aggregates all per-symbol signals received on a given date and produces a buy, sell, or hold recommendation per symbol. The *presentation layer* contains a portfolio simulator, which applies the recommendations at the daily closing price, maintains cash and positions net of commissions and slippage, and records the equity curve; and a metrics calculator, which computes the risk-adjusted metrics reported in Section 5.1.6.

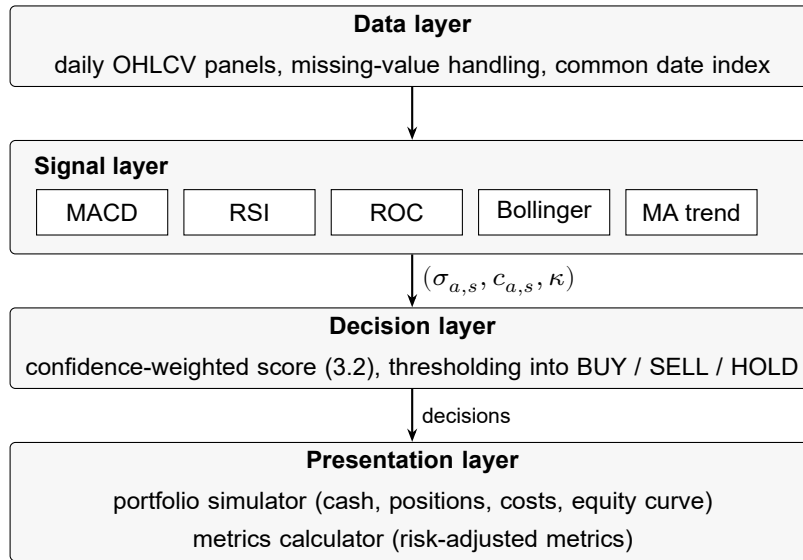


Fig. 3.1. System architecture. The four layers communicate through well-defined interfaces, so that new signal agents can be added without changing the decision layer.

All signal agents inherit from a common base class that guarantees a uniform interface. Following the principle of methodological consistency, each agent uses a single family of indicators with no mixed rules. This makes the aggregation argument of Section 3.1 well defined: each agent corresponds to a distinct forecaster f_k with its own bias-variance profile. The software realisation of these layers is described in Chapter 4.

3.5. Signal agents

Each signal agent acts as an independent and imperfect forecaster of the next-period directional move for a single symbol. To make the forecast-combination argument of Section 3.1 well defined, every agent is confined to a single methodological family, so that its forecast errors are as distinct as possible from those of the other agents.

For each agent, both its computation and its forecasting role are stated below, that is, the market behaviour it is designed to detect and the conditions under which its forecast tends to be wrong. In all cases, the agent returns a neutral (hold) signal when the history is too short to compute the indicator, and the confidence value is zero outside the active signal band.

MACD (moving-average convergence-divergence). The MACD line is the difference between a fast and a slow exponential moving average (EMA) of the closing price (default spans 8 and 17), and the signal line is an EMA of the MACD line (default span 9). The agent emits $+1$ on

an upward crossover of the two lines and -1 on a downward crossover, with confidence

$$\text{confidence} = \tanh\left(\frac{|h|}{2\sigma}\right), \quad (3.3)$$

where h is the current histogram value and σ its standard deviation over the last 50 bars. Role: as a momentum forecaster, the agent reports the persistence of recent directional moves; its forecast errors concentrate around trend reversals, where the crossover lags the turning point. Methodological family: momentum.

RSI (relative strength index). The agent computes the average gain and the average loss over a rolling window (default 14 days), and defines $\text{RSI} = 100 - 100/(1 + \text{RS})$ with $\text{RS} = \text{avg_gain}/\text{avg_loss}$. The action rule is

$$\text{action} = \begin{cases} \text{buy,} & \text{RSI} < 30, \\ \text{sell,} & \text{RSI} > 70, \\ \text{hold,} & \text{otherwise,} \end{cases} \quad (3.4)$$

with confidence equal to the normalised distance of the RSI from the nearest threshold. Role: as a mean-reversion forecaster, the agent bets against recent price extremes; its forecast errors are largest in strong trends, where the index can remain overbought or oversold for extended periods. Methodological family: mean-reversion.

ROC (rate of change). The rate of change is the percentage change in the closing price over a fixed number of days (default 10). The action rule is

$$\text{action} = \begin{cases} \text{buy,} & \text{ROC} > \tau, \\ \text{sell,} & \text{ROC} < -\tau, \\ \text{hold,} & \text{otherwise,} \end{cases} \quad (3.5)$$

where τ is a configurable threshold (default 0.02). Confidence equals $|\text{ROC}|$, clipped to $[0, 1]$. Role: as a momentum forecaster on a shorter horizon than MACD, the agent responds to the raw speed of price change; its forecast errors arise when a fast move reverses abruptly. Methodological family: momentum.

Bollinger Bands. The agent computes a simple moving average (SMA) of the closing price over a window (default 20) as the middle band, and upper and lower bands $U = \text{middle} + m\sigma$, $L = \text{middle} - m\sigma$, where m is a configurable multiplier (default 2). The action rule is

$$\text{action} = \begin{cases} \text{buy,} & \text{close} < L, \\ \text{sell,} & \text{close} > U, \\ \text{hold,} & \text{otherwise.} \end{cases} \quad (3.6)$$

Confidence is the normalised distance of the close from the nearest band. Role: as a volatility forecaster, the agent measures how far the price has stretched from its mean in units of the standard deviation and signals a reversion when the close pierces a band; its forecast errors occur when a band breakout develops into a sustained move instead of reverting. Because the bands respond to dispersion rather than to direction, the agent carries the least directional information of the five, which is reflected in its family weight (Section 3.6). Methodological family: volatility.

MA trend. The agent uses two SMAs: fast (default 50) and slow (default 200). A buy signal is emitted when the slow SMA exceeds the fast SMA, and a sell signal is emitted in the

opposite case. Confidence equals the absolute difference between the two SMAs divided by the latest close. Role: as a long-horizon trend forecaster, the agent identifies the prevailing market regime from the relation between a short and a long average; its forecast errors occur in choppy, trendless markets where the two averages cross repeatedly. Methodological family: trend.

3.6. Decision agent

The decision agent receives the list of signals produced on a given date and groups them by symbol. Signals with confidence below a global minimum confidence $c_{\min}$ are discarded. The remaining signals are aggregated into the score defined in (3.2). The family weights w_{κ} are fixed at

$$\begin{aligned} w_{\text{trend}} &= 1.3, & w_{\text{momentum}} &= 1.0, \\ w_{\text{mean-rev.}} &= 0.8, & w_{\text{volatility}} &= 0.5. \end{aligned} \quad (3.7)$$

The weights encode a fixed prior on the relative directional reliability of each methodological family, and their ordering follows directly from the empirical literature rather than from any in-sample search. Trend-following receives the largest weight because time-series momentum is among the most robust and persistent return predictors documented across equity markets and horizons [22, 21], and daily index constituents such as the WIG20 blue chips tend to exhibit sustained directional regimes. Momentum on a shorter horizon receives a unit weight as the reference family. Mean reversion is weighted below momentum because overbought and oversold rules are informative mainly in range-bound phases and are frequently wrong during strong trends. Volatility receives the smallest weight because band-based signals respond to dispersion rather than to direction, and therefore carry the least directional information of the four families.

The four levels are intentionally coarse, equally spaced multiples that keep the weights interpretable and the degrees of freedom low. They are held fixed across every symbol, regime, and evaluation window, and are never tuned on the test data. Retaining a fixed prior instead of estimating the weights is additionally supported by the forecast-combination literature, in which simple fixed weights are notoriously difficult to beat with estimated ones, because weight-estimation error tends to consume the gains it promises. That phenomenon is known as the *forecast combination puzzle* [43, 44].

The sensitivity of the results to this fixed scheme is examined empirically in Section 5.3.2, where each family weight is perturbed by up to $\pm 50\%$ and the whole vector is replaced by equal, inverted, in-sample-optimised, and inverse-variance-estimated alternatives. The leave-one-agent-out ablation in Section 5.3.1 covers the limiting case of removing an agent entirely, and regime-conditional weighting is discussed as future work in Section 6.2.

The score is then compared against mode-dependent thresholds to produce the final decision:

$$\text{action} = \begin{cases} \text{BUY}, & \text{score} \geq \theta_{\text{buy}}, \\ \text{SELL}, & \text{score} \leq \theta_{\text{sell}}, \\ \text{HOLD}, & \text{otherwise.} \end{cases} \quad (3.8)$$

The thresholds are $\theta_{\text{buy}} = -\theta_{\text{sell}} = 1.2$ in *passive* mode, 0.7 in *balanced* mode, and 0.3 in *aggressive* mode. In *passive* mode the decision agent additionally requires the sign of the score to agree with the dominant trend direction, which is defined as the sign-weighted sum of confidences of the trend-family agents.

The decision-agent parameters are the mode $m \in \{\text{passive}, \text{balanced}, \text{aggressive}\}$ and the minimum confidence threshold $c_{\min}$. These are the only parameters tuned during walk-forward

evaluation. All indicator hyperparameters remain fixed at their literature-conventional defaults throughout.

3.7. Agent taxonomy

Table 3.1 summarises the five signal agents along three dimensions: methodological family, signal type (crossover versus threshold), and the horizon of the underlying indicator. The forecasting rationale of each family is given with the agent descriptions in Section 3.5.

The taxonomy makes the heterogeneity of the ensemble explicit and motivates the family-level weights w_{κ} : agents that are expected to produce the most distinct forecast errors are pooled across families with different weights, consistent with the variance-reduction argument of Section 3.1. The empirical correlations of the agent signals, reported in Section 5.2.7, support the taxonomy: cross-family pairs, for example RSI versus ROC, or Bollinger versus ROC, exhibit the largest absolute correlations with opposing signs, consistent with their opposite interpretations of the same raw price data.

Table. 3.1. Taxonomy of the five signal agents. Horizon refers to the nominal lookback of the underlying indicator.

Agent	Family	Signal type	Horizon
MACD	momentum	crossover	17 d
RSI	mean-reversion	threshold	14 d
ROC	momentum	threshold	10 d
Bollinger	volatility	threshold	20 d
MA trend	trend	crossover	200 d

3.8. Chapter summary

Forecast-combination theory predicts that pooling imperfectly correlated forecasters reduces the variance of the combined forecast, and that the reduction is largest when the pooled forecasters make errors of different kinds. The system described in this chapter is built to instantiate that prediction: five agents, each confined to a single methodological family, feed a decision agent that combines their signals into the linear, confidence-weighted score of (3.2) and thresholds it into a trading action.

Two design choices deserve emphasis, because the evaluation returns to both. The family weights are a fixed prior taken from the empirical literature rather than an estimated quantity, which trades potential in-sample fit for robustness. The confidence values are heuristics rather than calibrated probabilities, which means the inverse-variance reading of the score is an analogy to be tested, not an established property. Chapter 5 tests both, along with the three hypotheses stated in Section 3.3.

4. IMPLEMENTATION

Chapter 3 described the system as a set of interacting abstractions: agents that emit signals, a decision agent that combines them, a simulator that executes the resulting orders. This chapter describes the software that realises those abstractions. It states the requirements the implementation had to satisfy, justifies the selected technologies, presents the structure of the code in UML form, walks through the execution of a single trading day, describes the preparation of the price data, catalogues the experiment harness that produces the results of Chapter 5, and explains how the implementation was verified.

The complete system comprises 46 Python modules, of which 16 are experiment scripts.

4.1. Requirements

The requirements were derived from the aim stated in Section 1.2 and from the evaluation protocol the system had to support. They are stated here in the form in which they governed the implementation.

4.1.1. Functional requirements

Table 4.1 lists the functional requirements. Requirements F1 to F4 follow from the system design of Chapter 3; F5 to F8 follow from the evaluation protocol, which needs to run the same pipeline hundreds of times under varying configurations.

Table. 4.1. Functional requirements of the system.

ID	Requirement
F1	Load daily OHLCV data for a configurable roster of symbols from local CSV files.
F2	Compute, for each agent, symbol and date, a directional signal, a confidence value and a methodological family, using only data available up to that date.
F3	Aggregate all signals for a symbol on a date into a single buy, sell or hold decision through the confidence-weighted score of (3.2).
F4	Simulate the execution of decisions on a cash portfolio, charging commission, slippage and, where applicable, a one-way transaction tax.
F5	Compute the risk-adjusted metrics of Section 5.1.6 from the resulting equity curve and trade list.
F6	Run an anchored walk-forward search over decision-agent parameters and evaluate the selected configuration on a held-out window.
F7	Run ablation, weight-sensitivity, cost-sensitivity, regime, bootstrap and confidence-validation experiments over the same pipeline.
F8	Export every experiment result to CSV so that the numbers reported in the thesis are traceable to a file.

4.1.2. Non-functional requirements and design constraints

The non-functional requirements are listed in Table 4.2. Two of them deserve comment, because they shaped the code more than the others.

Requirement N1, the absence of look-ahead, is a correctness property rather than a preference: a single leak invalidates the entire evaluation. It is enforced structurally rather than by

convention. Each agent receives a data frame already truncated at the current date, so an agent cannot observe a future bar even if its implementation were careless.

Requirement N3, the separation of signal computation from portfolio accounting, began as a performance concern and became an architectural principle. The experiment battery re-runs the backtest across grid points, ablations, weight perturbations and cost configurations, and recomputing indicator values on every run made the full battery impractically slow. Precomputing the signal table once and passing it into each backtest reduced the end-to-end runtime by more than an order of magnitude.

Table 4.2. Non-functional requirements and design constraints.

ID	Requirement
N1	No look-ahead: a signal for date t must depend only on data up to and including t .
N2	Determinism: repeated runs on identical inputs must produce identical outputs, so that reported numbers are reproducible.
N3	Separation of signal computation from portfolio accounting, so that cost and parameter sweeps do not recompute indicators.
N4	Extensibility: a new signal agent must be addable without modifying the decision layer or the simulator.
N5	Configurability: symbols, data paths, costs, decision parameters and agent hyperparameters must be declared outside the code.
N6	Market portability: switching to another exchange must require only a new data source and cost calibration.
C1	Every signal agent inherits from BaseAgent and implements a single method.
C2	Every signal agent uses one internally consistent methodology; mixed-indicator agents are not permitted.
C3	Signal agents may not make portfolio-level decisions; they only emit signals.
C4	Every signal carries a confidence value alongside its direction.

Constraints C1 to C4 are architectural rules adopted at the start of the project. C2 is the one with theoretical weight: it is what makes the forecast-combination argument of Section 3.1 well defined, because an agent mixing several indicator families would not correspond to a single forecaster with an identifiable error profile.

4.2. Technology stack

The system is implemented in Python 3.11. Python was selected because the numerical and machine-learning libraries required by the evaluation are mature and interoperable in that ecosystem, and because the resulting code is readable enough to serve as documentation of the method, which matters for a thesis whose contribution is partly methodological.

Table 4.3 lists the libraries and the role of each. The dependency set is deliberately small: five libraries carry the entire multi-agent system, and the remaining two are needed only for the machine-learning baselines. That asymmetry is itself a result. The ensemble requires no training infrastructure at all, which is the deployment argument made in Section 5.4.7.

Two choices merit justification. Rolling-window indicator computation is delegated to pandas rather than implemented manually, because the library implementations are well tested and their treatment of leading undefined values is explicit, which matters for N1. LightGBM was chosen for the second machine-learning baseline in preference to another neural architecture, because a

Table. 4.3. Technologies used in the implementation, with the role of each.

Component	Version	Role
Python	3.11	Implementation language.
pandas	≥ 2.0	Price panels, rolling-window indicator computation, equity curves.
NumPy	≥ 1.24	Vectorised numerics underlying the metric calculations.
Matplotlib	≥ 3.7	Generation of the figures reported in Chapter 5.
PyYAML	≥ 6.0	Declarative configuration of symbols, costs, agents and decision parameters.
SciPy	≥ 1.10	Spearman rank correlation used in the confidence validation of Section 5.2.8.
PyTorch	≥ 2.2	LSTM directional baseline of Section 5.1.5.
LightGBM	≥ 4.0	Gradient-boosted-trees baseline of Section 5.1.5.

gradient-boosted tree ensemble is structurally different from a recurrent network, so agreement between the two baselines is more informative than agreement between two variants of the same model family.

4.3. Software architecture

The four conceptual layers of Section 3.4 map onto Python packages as follows. The data layer is a loader function operating on a directory of CSV files. The signal layer is the package `src/agents`, containing the five agents and their common base. The decision layer is a single class, `DecisionAgent`. The presentation layer is the package `src/utils`, containing the portfolio simulator, the metrics calculator, and the plotting helpers. The experiment harness in `scripts/sits` above all four.

Fig. 4.1 shows the class structure of the signal and decision layers.

4.3.1. The agent contract

The base class is deliberately minimal. It declares two class attributes, `name` and `kind`, the latter being the methodological family $\kappa(a)$ of Section 3.2; it provides two helpers that construct the return dictionary; and it declares a single abstract method that every agent must implement. Listing 4.1 reproduces it in full.

Three properties of this contract are worth noting. The `_pack` helper clips confidence into $[0, 1]$, so no agent can inflate its own weight in the aggregate score by returning an out-of-range value. The `_neutral_signal` helper gives every agent a uniform way to abstain, which agents use when the price history is too short to compute their indicator. And because the only input is a data frame and a symbol, the interface says nothing about portfolio state, which enforces constraint C3 structurally.

Adding a new agent therefore means writing one class with one method, and registering it in the configuration file. Neither the decision agent nor the simulator changes, which satisfies N4.

4.3.2. The decision layer

`DecisionAgent` implements (3.2) and the thresholding rule (3.8) directly. Its `decide` method groups the signals of a date by symbol, delegates to a per-symbol routine, discards hold decisions, and applies an optional cap on the number of simultaneous positions. Listing 4.2 shows the per-symbol routine, which is the concrete realisation of the score.

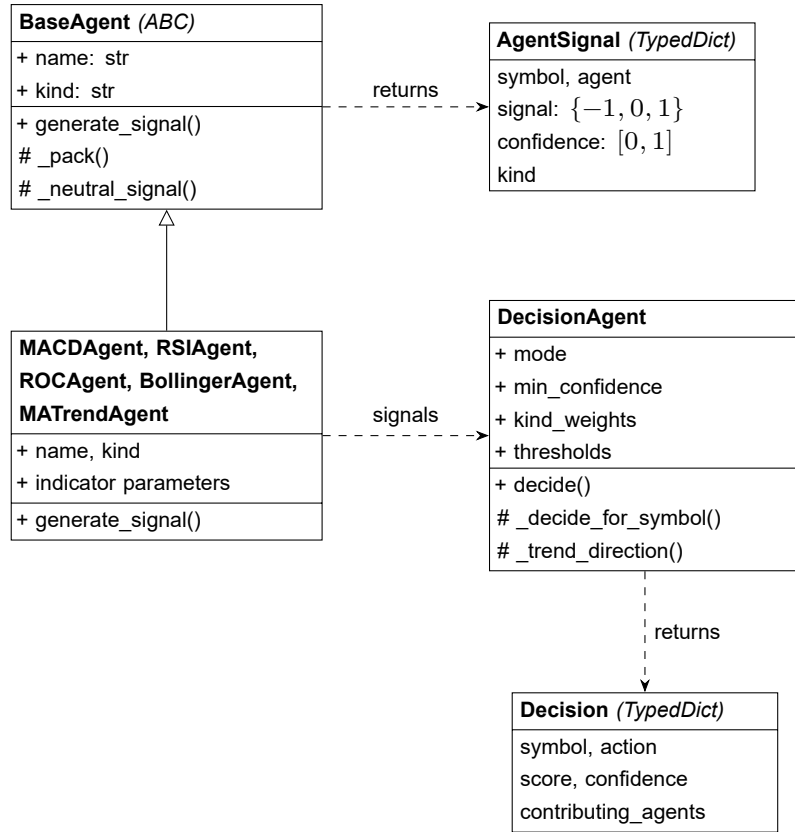


Fig. 4.1. Class structure of the signal and decision layers. All five signal agents inherit from the abstract base class and communicate only through the `AgentSignal` dictionary, so the decision layer is independent of how any individual signal is computed.

The correspondence with the theory is close enough to read line by line. The filter on `min_confidence` is the threshold $c_{\min}$; the product of signal, confidence and family weight is one term of the sum in (3.2); and the comparison against the mode-dependent thresholds is (3.8). The additional branch under *passive* mode implements the trend-agreement requirement described in Section 3.6.

One implementation detail carries analytical value. Every decision records the list of contributing agents together with their individual signals and confidences. This is what makes the per-agent attribution described in Section 5.4.6 possible: the score of any day can be decomposed after the fact into the contribution of each agent, without re-running the pipeline.

4.4. Implementation of the trading pipeline

Fig. 4.2 shows the execution of a single trading day. The loop over dates is driven by the `backtest` function, which holds the precomputed signal table and calls the decision agent and the simulator in turn.

4.4.1. Signal precomputation and the absence of look-ahead

The signal table is built once by iterating over the dates common to all symbols and, for each date and symbol, calling every agent with the price history truncated at that date. The truncation is the mechanism that enforces N1: an agent is handed `df.loc[:date]` and therefore cannot observe a future bar.

This construction is safe even though it uses the full price history, because the signal for

```

1 from abc import ABC, abstractmethod
2 import pandas as pd
3 from src.agents.base.AgentSignal import AgentSignal
4
5
6 class BaseAgent(ABC):
7     name: str
8     kind: str
9
10    def _neutral_signal(self, symbol: str) -> AgentSignal:
11        return {
12            "symbol": symbol,
13            "agent": self.name,
14            "signal": 0,
15            "confidence": 0,
16            "kind": self.kind
17        }
18
19    def _pack(self, symbol: str, signal: int, confidence: float) -> AgentSignal:
20        return {
21            "symbol": symbol,
22            "agent": self.name,
23            "signal": signal,
24            "confidence": min(max(confidence, 0.0), 1.0),
25            "kind": self.kind
26        }
27
28    @abstractmethod
29    def generate_signal(self, data: pd.DataFrame, symbol: str) -> AgentSignal:
30        raise NotImplementedError

```

Listing 4.1. The abstract base class defining the agent contract.

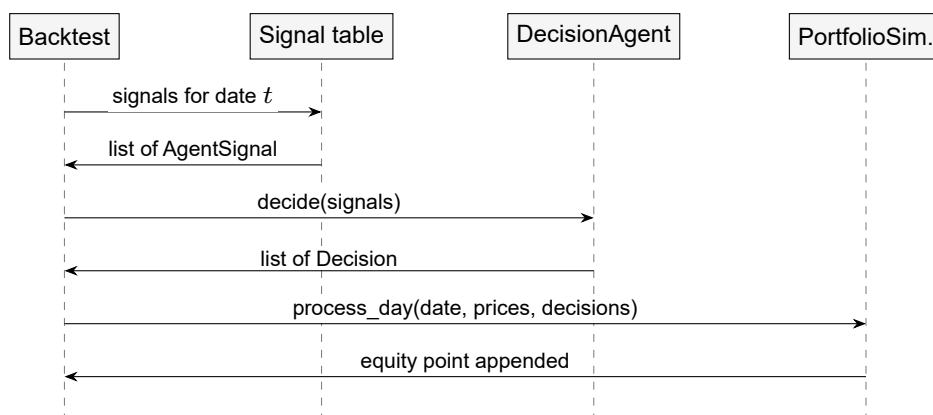


Fig. 4.2. Execution of one trading day. Signals are read from the precomputed table rather than recomputed, which is what makes the parameter and cost sweeps of Chapter 5 affordable.

```

1  def _decide_for_symbol(self, symbol: str, signals: List[dict]) -> Decision:
2      filtered = [
3          s for s in signals
4          if s["signal"] != 0 and s["confidence"] >= self.min_confidence
5      ]
6
7      if not filtered:
8          return self._hold(symbol)
9
10     trend_dir = self._trend_direction(signals)
11     score = 0.0
12     contributors = []
13
14     for s in filtered:
15         if self.mode == "passive" and trend_dir != 0:
16             if s["signal"] != trend_dir:
17                 continue
18
19         weight = self.kind_weights.get(s["kind"], 1.0)
20         contribution = s["signal"] * s["confidence"] * weight
21         score += contribution
22         contributors.append({
23             "agent": s["agent"],
24             "signal": s["signal"],
25             "confidence": s["confidence"],
26             "kind": s["kind"],
27         })
28
29     thresholds = self.thresholds[self.mode]
30
31     if score >= thresholds["buy"]:
32         return self._decision(symbol, "BUY", score, contributors)
33     elif score <= thresholds["sell"]:
34         return self._decision(symbol, "SELL", score, contributors)
35     else:
36         return self._hold(symbol)

```

Listing 4.2. Per-symbol aggregation: the implementation of the confidence-weighted score of (3.2) and of the thresholding rule of (3.8).

date t depends only on data up to t by construction. Querying the resulting table over a later sub-window therefore introduces no leakage. That property is what allows the same table to serve the training window, the test window, every ablation and every cost configuration.

4.4.2. Portfolio accounting

The simulator maintains cash, a dictionary of open positions, a trade log and an equity curve. On each day it processes sell decisions before buy decisions, so that capital released by a closing position is available to a new one on the same date, and then records the mark-to-market value of the portfolio.

The cost model of Section 5.1.2 is implemented in the fill computation. A purchase fills at $p(1 + s)$ and a sale at $p(1 - s)$, where s is the one-way slippage rate, so the round trip pays slippage twice. Commission is charged on both legs. A one-way transaction tax, used for the UK Stamp Duty Reserve Tax in Section 5.3.7, is charged on purchases only and is kept as a separate

rate so that the two legs can carry asymmetric charges.

Position sizing accounts for entry charges rather than ignoring them. The allocation for a new position is a fixed fraction of current cash, and the affordable notional is that allocation divided by $1 + \gamma + \tau$, where γ is the commission rate and τ the buy-side tax rate. The share count is then the integer floor of the affordable notional divided by the fill price, so the simulator never buys fractional shares and never spends cash it does not hold. Listing 4.3 shows the routine.

```

1      # Size the position against cash capacity inclusive of entry charges
2      allocation = self.cash * self.position_size
3      max_spend = allocation / (1.0 + buy_charge_rate)
4      if max_spend <= 0:
5          return
6
7
8      quantity = int(max_spend // fill_price)
9      if quantity <= 0:
10         return
11
12     notional = quantity * fill_price
13     entry_charge = notional * buy_charge_rate
14     total_cost = notional + entry_charge
15     if total_cost > self.cash:
16         return
17
18     self.cash -= total_cost
19
20     self.positions[symbol] = {
21         "symbol": symbol,
22         "quantity": quantity,
23         "entry_price": fill_price,
24         "entry_date": date,
25         "entry_commission": entry_charge,
26     }
27
28     self.trades.append({
29         "symbol": symbol,
30         "action": "BUY",
31         "price": fill_price,
32         "quantity": quantity,
33         "date": date,
34         "pnl": None,
35     })
36
37     def _sell(self, symbol: str, price: Optional[float], date: pd.Timestamp) -> None:
38         if price is None:
39             return
40         if symbol not in self.positions:
41             return

```

Listing 4.3. Entry-side accounting in the portfolio simulator, including slippage, commission and the one-way buy tax.

Realised profit on a closed trade is computed net of both commissions and of the round-trip slippage already embedded in the two fill prices, which is why the entry commission is stored with the position rather than discarded after the purchase.

4.4.3. Metrics

`MetricsCalculator` consumes the equity curve and the trade log and implements the definitions of Section 5.1.6. Annualisation uses $\sqrt{252}$ for ratio statistics and calendar years, computed as elapsed days divided by 365.25, for CAGR. Each metric guards against the degenerate inputs that arise during the parameter sweep, in particular an empty trade list and a flat equity curve, and returns zero rather than raising, so that a configuration that never trades is comparable rather than fatal. This is the behaviour that makes the *passive* column of Table 5.1 well defined.

4.5. Data preparation

Price data are daily OHLCV series downloaded from Stooq [45] as one CSV file per instrument, with columns Date, Open, High, Low, Close and Volume. Files are named by the lower-case ticker, and the loader skips any symbol declared in the configuration that has no corresponding file, so a partial data directory degrades gracefully rather than failing.

Three preparation steps are applied. First, all series are truncated at the common end date of 2026/03/31, so that no instrument contributes observations the others cannot. Second, rows with missing values are dropped per symbol. Third, the backtest intersects the date indices of all loaded symbols and iterates only over dates common to every instrument, which guarantees that the portfolio sees a consistent cross-section on every date.

The WIG20 roster comprises nineteen constituents and the index series. Žabka Group SA. is excluded because its listing on 2024/10/17 falls inside the test window, so including it would introduce a constituent whose history is absent from the training window. The FTSE 100 panel is prepared identically, with 20 constituents and the index series, and is held in a separate directory selected through the configuration file.

4.6. Experiment harness

Each experiment reported in Chapter 5 is produced by a dedicated script that loads the configuration, builds the agents, runs the relevant sweep and writes a CSV file. Table 4.4 maps the experiments to their scripts and outputs.

All runtime parameters live in a YAML file rather than in code, which satisfies N5. The configuration declares the data directory, the symbol roster, initial cash and position size, the commission and slippage rates, the anchored split date and fold count, the decision-agent mode and minimum confidence, the enabled agents with their indicator parameters, and the hyperparameter grids of the two machine-learning baselines. Disabling an agent is a matter of removing one line.

Market portability, requirement N6, is realised through this file. The FTSE 100 replication of Section 5.3.7 uses a second configuration that differs from the primary one only in the data directory, the symbol roster and the cost calibration. No code path is specific to either market.

4.7. Verification of the implementation

Because the contribution of this work rests on the credibility of its numbers, verification was directed at the properties that would invalidate them if violated, rather than at line coverage.

Absence of look-ahead. The critical property, N1, is verified structurally rather than by test. Agents receive a truncated data frame and have no access to the full panel, so a leak would require an agent to reach outside its argument. The precomputation step preserves this property because a signal at date t is a function of data up to t only, which makes the table equivalent to computing

Table. 4.4. Experiment harness: the script that produces each result reported in Chapter 5, and the file it writes.

Experiment	Script	Output
Walk-forward selection and headline metrics	run_walk_forward.py	walk_forward_*.csv
Leave-one-agent-out ablation	run_ablation.py	ablation.csv
Cost-sensitivity sweep	run_cost_sensitivity.py	cost_sensitivity.csv
Risk-adjusted and volatility-matched metrics	run_risk_metrics.py	risk_metrics*.csv
Market-regime decomposition	run_regime_analysis.py	regime_analysis.csv
Bootstrap intervals and paired tests	run_stats.py	stats.csv
Weight sensitivity	run_weight_sensitivity.py	weight_sensitivity.csv
Confidence validation	run_confidence_validation.py	confidence_validation_*.csv
Selection-objective comparison	run_objective_comparison.py	objective_comparison.csv
LSTM baseline	run_ml_baselines.py	ml_baseline*.csv
GBM baseline	run_gbm_baseline.py	gbm_baseline*.csv
Figures	regenerate_figures.py, regenerate_extra_figures.py	img/*.pdf

signals date by date inside each backtest. This equivalence was checked directly by running a backtest with precomputed signals and one that recomputes signals per date, and confirming that the two produce identical equity curves.

Determinism. Requirement N2 is verified by re-running each experiment script and comparing the regenerated CSV against the committed one. The multi-agent pipeline contains no stochastic component, so agreement is exact. The two machine-learning baselines are seeded explicitly and average three seeded models, so their outputs are reproducible given the same library versions.

Consistency of the accounting. The portfolio simulator was checked against invariants that must hold for any execution path: cash never becomes negative, because a purchase is rejected when its total cost exceeds available cash; the share count is integral; a position is never opened twice for the same symbol; and the realised profit reported for a closed trade equals the difference in fill prices times the quantity, less both commissions.

Cross-validation of the metrics. The metric implementations were verified against independently computed values on the same equity curves, and against the relationships that must hold between them, in particular that the Calmar ratio equals CAGR divided by the absolute maximum drawdown, and that the Sharpe and Sortino ratios coincide when no daily return is negative.

Verification of the results. The strongest verification of the implementation is the evaluation itself. The battery reported in Chapter 5 exercises the same code paths under hundreds of configurations, and the results move in the directions that the design predicts: removing an agent degrades performance modestly rather than catastrophically, raising costs lowers returns monotonically, and the regime decomposition sums back to the aggregate figures. A defect in the aggregation or accounting would be expected to break at least one of these regularities.

The limitation of this approach should be stated plainly: the implementation carries no automated unit-test suite, so the verification described above establishes the properties that matter for the reported results rather than the correctness of every function in isolation. Adding such a suite is a natural piece of further work.

4.8. Chapter summary

The implementation is small, and deliberately so. Five agents, one decision class, one simulator and one metrics calculator carry the entire system, and the dependency set for that system comprises five widely used libraries with no training infrastructure at all. The complexity that exists lives in the experiment harness rather than in the system under study, which is the appropriate distribution for work whose main claim is about evaluation.

Two implementation decisions had consequences beyond the code. Enforcing one methodological family per agent, constraint C2, is what makes the forecast-combination argument well defined rather than merely suggestive. Separating signal computation from portfolio accounting, requirement N3, is what made the sensitivity analyses of Chapter 5 affordable, and therefore what made the robustness claims possible at all.

5. EVALUATION METHODOLOGY AND RESULTS

This chapter reports the empirical part of the thesis. Section 5.1 describes the evaluation protocol: the datasets, the transaction-cost model, the anchored walk-forward procedure, the baselines, the risk-adjusted metrics, the regime labelling, the volatility-matched comparison, the cost-sensitivity sweep, and the bootstrap tests. Sections 5.2 and 5.3 report the results on the WIG20 panel and the robustness checks together with the external validation on the FTSE 100. Section 5.4 interprets the evidence against the three hypotheses of Section 3.3, and Section 5.5 closes the chapter.

The protocol described here is the element of the work that addresses the four methodological weaknesses identified in Chapter 2, and the stage that the template calls testing and verification of results: every headline number is produced on data that was never used to select a parameter, net of realistic frictions, accompanied by confidence intervals, and stress-tested against ablations, weight perturbations, cost grids, and a second market.

5.1. *Experimental setup*

5.1.1. *Dataset*

Both experiments use daily OHLCV panels from the public repository Stooq [45]. On each market, the ensemble, the buy-and-hold index benchmark, and the machine-learning baselines trade the same fixed roster of constituents; incomplete rows are dropped per symbol.

The primary panel covers the WIG20 index and nineteen WIG20 stocks on the Warsaw Stock Exchange, excluding Żabka Group SA., whose listing on 2024/10/17 is much more recent than that of the remaining constituents. The period covers 2021/05/26 to 2026/03/31, roughly 1,220 trading days. Values are denominated in PLN.

External validation uses 20 liquid FTSE 100 constituents and the FTSE 100 index on the London Stock Exchange over the same calendar window in GBP. The anchored split is identical on both markets, with training through 2024/06/30 and testing from 2024/07/01 through 2026/03/31. Agents, hyperparameters, and evaluation procedures are unchanged; only the price panels, index series, and transaction-cost calibration differ (Sections 5.1.2 and 5.3.7). The preparation of both panels is described in Section 4.5.

5.1.2. *Transaction-cost model*

All backtests include a proportional commission of 39 basis points (bps) of notional per trade and a one-way slippage of 5 bps applied to the executed price. These values are in the typical range of Polish retail brokerages for WSE-listed equities. The commission is deducted from cash both on entry and on exit. Slippage raises the effective buy price and lowers the effective sell price by the quoted amount. Realised profit or loss is reported net of the commissions on both legs and of the round-trip slippage.

This choice is conservative for a retail investor and significantly reduces the headline gross return of high-turnover configurations, which is one of the methodological concerns raised in the literature on backtest reliability [12]. A full sensitivity sweep over commission and slippage is reported in Section 5.2.6.

5.1.3. Walk-forward protocol

The evaluation follows an anchored walk-forward protocol. The full period is split into a training window that spans 2021/05/26 to 2024/06/30 and a held-out test window that spans 2024/07/01 to 2026/03/31. On the training window a grid search is performed over the decision-agent parameters

$$\begin{aligned} \text{mode} &\in \{\text{passive}, \text{balanced}, \text{aggressive}\}, \\ c_{\min} &\in \{0, 0.05, 0.10, \dots, 0.30\}, \end{aligned} \tag{5.1}$$

and the configuration that maximises the in-sample total return is selected. The selected configuration is then evaluated untouched on the test window. The indicator hyperparameters are not part of the grid and remain fixed throughout. This design keeps the number of degrees of freedom low and matches the way technical-analysis rules are treated in most of the cited literature.

Total return is deliberately retained as the selection objective, even though the strategy is evaluated primarily on risk-adjusted metrics. The reason is statistical rather than conceptual. On a low-dimensional grid of 21 configurations, in-sample total return is a simple, low-variance ranking criterion, whereas ratio statistics such as the Sharpe ratio carry substantial estimation error on samples of this length [46] and become degenerate for configurations that trade rarely. The *passive* mode produces no trades at any threshold (Table 5.1), so its in-sample Sharpe ratio is undefined rather than merely low, and a risk-adjusted objective would have to break such ties arbitrarily. To verify that this choice is not load-bearing, Section 5.3.6 re-runs the selection under Sharpe-ratio and Calmar-ratio objectives on the identical grid and shows that the selected configuration, and therefore every out-of-sample number in the thesis, is unchanged on both markets.

As a robustness check, the results of an anchored, expanding walk-forward with three folds are additionally reported. For each fold $k \in \{1, 2, 3\}$, the test slice is one of three roughly equal chronological slices of the held-out window, and the training window spans 2021/05/26 up to the day before the test slice begins. A fresh grid search is performed on each fold and the winning configuration is evaluated on the test slice of that fold. The objective of this exercise is to show that the headline configuration is not an artefact of a particular single split.

Algorithm 1 summarises the full protocol in pseudocode. Its key feature is that signals are precomputed once outside all grid-search and ablation loops. This decouples the decision-agent grid search from signal computation, reduces the end-to-end runtime by more than an order of magnitude, and makes the cost-sensitivity sweep of Section 5.1.9 cheap, because changing costs does not require recomputing any agent signal.

5.1.4. Baselines

Three families of baselines are used. The first is *index buy-and-hold*: on WIG20 the benchmark invests the initial cash into the WIG20 index on the first date of the test window and holds the position until the end; on the FTSE replication the same rule applies to the FTSE 100 index. Buy-and-hold is a standard benchmark in the evaluation of trading strategies [14].

The second family is a set of *single-agent* runs in which the decision agent receives signals from only one signal agent, which produces five baselines, one per indicator. In addition, five *leave-one-agent-out* configurations are run, each of which removes one agent from the ensemble. The third family comprises two *machine-learning baselines*, described in Section 5.1.5.

All baselines are evaluated on the same test window on each market, use the market-specific transaction costs described in Sections 5.1.2 and 5.3.7, and adopt the same decision-agent

Algorithm 1 Anchored walk-forward with precomputed signals

```
1: Input: prices  $P$ ; agent set  $\mathcal{A}$ ; modes  $\mathcal{M}$ ; thresholds  $\mathcal{C}$ ; train/test split  $(T_{\text{tr}}, T_{\text{te}})$ ; costs  $(\gamma, s)$ .  
2: Output: selected  $(m^*, c^*)$ ; test-window metrics.  
3:  $S \leftarrow \text{PrecomputeSignals}(P, \mathcal{A})$  {all dates, all symbols}  
4:  $\text{best} \leftarrow -\infty$   
5: for all  $m \in \mathcal{M}, c \in \mathcal{C}$  do  
6:    $E_{\text{tr}} \leftarrow \text{Backtest}(S, T_{\text{tr}}, m, c, \gamma, s)$   
7:    $r \leftarrow \text{TotalReturn}(E_{\text{tr}})$   
8:   if  $r > \text{best}$  then  
9:      $\text{best} \leftarrow r; (m^*, c^*) \leftarrow (m, c)$   
10:  end if  
11: end for  
12:  $E_{\text{te}} \leftarrow \text{Backtest}(S, T_{\text{te}}, m^*, c^*, \gamma, s)$   
13: return  $(m^*, c^*), \text{Metrics}(E_{\text{te}})$ 
```

configuration selected by the walk-forward protocol. The only exceptions are buy-and-hold, which has no decision-agent parameters, and the machine-learning baselines, which replace the decision agent with a learned classifier.

5.1.5. Machine-learning baselines

As a modern learning-based reference point, an LSTM directional classifier in the style of [4] is implemented. For each symbol the model consumes a rolling window of causal per-day features derived from the same price history available to the agents: multi-horizon returns, realised volatility, and the normalised RSI, ROC, MACD-histogram, Bollinger %b, and moving-average-spread indicators. It predicts the probability that the h -day-ahead return is positive. The network is a single-layer LSTM followed by dropout and a linear output unit, trained with the Adam optimiser under binary cross-entropy.

To keep the comparison fair, all model-selection choices are made strictly in sample: the prediction horizon h , the operating threshold, and early stopping are selected on an internal validation slice carved chronologically from the training window, and the held-out test window is never used for any tuning. Feature standardisation statistics are estimated on the training window only, so the procedure carries no look-ahead.

Per-symbol probabilities are mapped to long or flat positions through a symmetric no-trade band around 0.5: a probability above $0.5 + \delta$ opens or holds a long position, a probability below $0.5 - \delta$ closes it, and intermediate probabilities leave the current position unchanged. This mirrors the hold behaviour of the decision agent and prevents the excessive turnover that would otherwise be induced by daily re-prediction. On both markets the in-sample selection returns a horizon of $h = 3$ days and a band of $\delta = 0.03$. The resulting long and flat signals are executed through the same portfolio simulator, under identical costs and position sizing, so that any difference reflects the decision rule rather than the accounting.

The second machine-learning baseline is a gradient-boosted-trees (GBM) directional classifier [26], implemented with LightGBM [27]. Gradient-boosted trees are consistently among the strongest learners on tabular financial features in large-scale empirical comparisons [28, 29], which makes them a natural, structurally different complement to the recurrent LSTM.

For each symbol and day the GBM model consumes the same causal per-day features

as the LSTM, augmented with lagged copies of each feature at lags of up to 20 trading days, so that the flat feature vector carries a temporal context comparable to the LSTM lookback window. Training pools feature rows across symbols, splits them chronologically into training and validation slices, applies early stopping on validation log-loss, and averages three differently seeded models. Exactly as for the LSTM, the prediction horizon and the no-trade band are selected strictly in sample by the net-of-cost Sharpe ratio of a validation-slice portfolio simulation, and the resulting long and flat signals are executed through the same portfolio simulator under identical costs. The in-sample selection returns $h = 1$ and $\delta = 0.03$ on WIG20, and $h = 3$ and $\delta = 0.03$ on the FTSE 100.

5.1.6. Risk-adjusted metrics

The following metrics are computed on the equity curve and on the list of closed trades. **Total return** is the percentage change in portfolio value from the first to the last date of the evaluation window. **CAGR** (compound annual growth rate) is the annualised return over the same window. **Max drawdown** is the largest peak-to-trough decline, reported as a negative fraction. **Sharpe ratio** [38] is the annualised ratio of excess return over standard deviation, using a factor of $\sqrt{252}$ and a risk-free rate of zero.

Sortino ratio [39] replaces the denominator with downside deviation, which penalises only below-target returns:

$$\text{Sortino} = \frac{\sqrt{252} \bar{r}}{\sqrt{\frac{1}{N} \sum_{t=1}^N \min(r_t, 0)^2}}, \quad (5.2)$$

where $\bar{r}$ is the mean daily return and the denominator is the downside deviation. **Calmar ratio** [40] is CAGR divided by the absolute value of the maximum drawdown. **Ulcer Index** [41] aggregates the full drawdown path:

$$\text{UI} = 100 \cdot \sqrt{\frac{1}{N} \sum_{t=1}^N \text{dd}_t^2}, \quad (5.3)$$

where dd_t is the running drawdown of the equity curve at date t . **Martin ratio** [41] is the annualised excess return divided by the Ulcer Index.

The Ulcer and Martin statistics are more sensitive to prolonged drawdowns than the peak-to-trough maximum, and they are used here as tie-breakers whenever two strategies have similar peak drawdowns but different drawdown profiles. **Number of trades** is the count of closed trades, and **win rate** is the fraction of closed trades with positive realised profit.

5.1.7. Market-regime labelling

To avoid comparing the ensemble and buy-and-hold only on pooled aggregates, each trading day of the test window is labelled as belonging to one of three regimes: *bull*, *correction*, or *sideways*. The labelling uses only information available up to date t from the WIG20 index itself, so it does not peek at future prices.

Let high_t be the rolling 252-day maximum of the WIG20 closing price at t , and let ret_t be the price relative $\text{close}_t / \text{high}_t$. A day is labelled *correction* when $\text{ret}_t < 1 - 0.10$, that is, the index is at least ten percent below its trailing one-year high. A day is labelled *bull* when $\text{ret}_t \geq 1 - 0.05$, that is, within five percent of the trailing high, and the 50-day SMA exceeds the 200-day SMA of the WIG20 index. All other days are labelled *sideways*.

These thresholds are standard in the practitioner literature on market regimes and are not tuned on the training data. Applying the labelling to the test window produces 52.87% bull days,

21.38% correction days, and 25.75% sideways days (Table 5.5), which means the test window covers all three regimes and the ensemble is evaluated under each of them.

5.1.8. Volatility-matched comparison

The volatility-matched comparison is an analytical normalisation, not a directly implementable trading result; its purpose is to place the ensemble and the benchmark on the same risk budget so that their returns can be compared on equal terms. The intuition is straightforward. Because the ensemble is out of the market on many days, it runs at a lower volatility than the fully invested benchmark, and a raw comparison of cumulative returns therefore penalises the ensemble for taking less risk. To remove this confound, the ensemble returns are multiplied by a single constant chosen so that the strategy and the benchmark have the same ex-ante volatility, after which the two equity curves are compared.

The formal construction is as follows. A standard criticism of the Sharpe ratio is that it is scale-invariant: it evaluates return per unit risk, but not the absolute level of risk [13]. A strategy with a Sharpe ratio of 1.0 and a realised annualised volatility of 10% produces a smaller cumulative return than a strategy with a Sharpe ratio of 1.0 and a realised volatility of 20%, simply because the second strategy takes more risk.

To remove this confound, the equity curve of the ensemble is scaled, ex ante, to the realised volatility of buy-and-hold WIG20 on the training window. The target volatility is the annualised standard deviation of WIG20 daily returns over the training window, $\sigma^{\text{target}} = \sqrt{252} \text{std}(r_{\text{train}}^{\text{WIG20}}) = 0.22$. The strategy volatility on the training window is $\sigma^{\text{strat}} = \sqrt{252} \text{std}(r_{\text{train}}^{\text{ensemble}}) = 0.11$. The leverage factor applied to the out-of-sample ensemble return is $\lambda = \sigma^{\text{target}} / \sigma^{\text{strat}} = 2.06$, constant over the test window. Because λ is computed purely from training-window data, the volatility-matched comparison is genuinely ex ante and does not peek at out-of-sample returns.

Two cautions apply to this comparison. First, the unleveraged ensemble remains the primary finding; the volatility-matched figures are a risk-budget counterfactual, not a recommendation to trade with leverage. Second, the required leverage factor of $\lambda = 2.06$ is above the level that most retail investors on the WSE can realistically access or would prudently use, so the volatility-matched results should be read as an analytical upper bound on the risk-normalised return of the strategy rather than as an attainable outcome. The scaled equity curve and its metrics are reported in Section 5.2.4.

5.1.9. Cost-sensitivity sweep

In addition to the baseline calibration of 39 bps commission and 5 bps slippage, the transaction-cost parameters are swept over a grid

$$\begin{aligned} \text{commission} &\in \{0, 10, 20, 39, 60, 80, 100\} \text{ bps}, \\ \text{slippage} &\in \{0, 5, 10, 20\} \text{ bps}, \end{aligned} \tag{5.4}$$

and the walk-forward backtest is re-run at each grid point. Crucially, the agent signals are precomputed once and passed into each backtest, so that the cost sweep changes only the portfolio-level accounting. The purpose of this experiment is to quantify how much of the risk-adjusted edge of the ensemble over buy-and-hold comes from the specific cost calibration. Section 5.2.6 reports the full grid.

5.1.10. Bootstrap confidence intervals and tests

In intuitive terms, a bootstrap repeatedly redraws the observed daily returns at random with replacement, recomputes the metric of interest on each redrawn series, and uses the spread of the resulting values to quantify how much the metric could have varied by chance. This procedure makes no assumption that returns follow a normal distribution.

Point estimates of total return and Sharpe ratio are accompanied by a 95% percentile bootstrap confidence interval. The bootstrap is computed on the vector of daily returns of the equity curve, with 10,000 resamples drawn with replacement. For the comparison against the local index buy-and-hold benchmark, against the best single-agent baseline, and against the machine-learning baselines of Section 5.1.5, a paired bootstrap test is performed: at each resample, the same set of dates is used for both series, the difference of the statistic of interest is computed, and the one-sided p -value reports the fraction of resamples with a non-positive difference. A p -value below 0.05 is treated as evidence that the strategy dominates the baseline on the test window.

The bootstrap is a natural choice here because the distribution of trading-strategy metrics is known to be non-Gaussian and path-dependent, and bootstrap methods have been advocated for this setting in prior work [15]. As is standard in the literature on trading-strategy evaluation, the bootstrap tests are not expected to be powered to detect moderate effect sizes on a single test window of just under two years [12]. The p -values are therefore reported alongside point estimates and confidence intervals rather than relied upon as the sole arbiter of performance.

5.2. Results on the WIG20 panel

This section reports the results obtained on the primary panel. It begins with the parameter selection performed on the training window, then reports the headline and risk-adjusted comparisons against buy-and-hold on the held-out window, decomposes them by market regime, and closes with two experiments that examine the internal mechanism of the ensemble: the empirical correlations between agent signals, and an audit of the confidence heuristic.

5.2.1. Parameter selection on training data

Table 5.1 reports the in-sample total return of every combination of decision-agent mode and minimum confidence on the training window. The *passive* mode rarely reaches its high buy and sell thresholds, and for every value of minimum confidence it produces no trades. The *balanced* mode trades moderately, with positive but small total return across all thresholds. The *aggressive* mode produces the highest total returns overall, and the best configuration on the training window is (*aggressive*, $c_{\min} = 0.05$), which is the configuration selected for out-of-sample evaluation.

The result is consistent with the well-known observation that moderate signal tolerance on a trend-rich daily dataset tends to maximise gross return, although net return is reduced noticeably by the transaction-cost model of Section 5.1.2.

5.2.2. Headline results on held-out data

Table 5.2 reports the headline metrics on the held-out test window for the selected configuration and for buy-and-hold WIG20. All numbers are net of the transaction costs described in Section 5.1.2.

On gross total return, buy-and-hold leads: it earns 0.31 over 435 trading days, whereas the ensemble earns 0.23 (CI $[-0.10, 0.68]$), and the one-sided bootstrap p -value for the hypothesis that the ensemble exceeds buy-and-hold on total return is 0.636.

Table. 5.1. In-sample total return (train window, 2021-2024) by decision-agent mode and minimum confidence. Higher is better. Selected configuration in bold.

$c_{\min}$	passive	balanced	aggressive
0.00	0.00	0.05	0.48
0.05	0.00	0.09	0.55
0.10	0.00	0.09	0.49
0.15	0.00	0.11	0.40
0.20	0.00	0.23	0.28
0.25	0.00	0.16	0.25
0.30	0.00	0.28	0.23

Once the comparison moves to the risk axis, the picture reverses. The Sharpe ratio of the ensemble is 1.06 (CI $[-0.43, 2.57]$), compared with 0.82 for buy-and-hold, and the p -value on Sharpe is 0.297. The maximum drawdown is -0.06 against -0.17 , a reduction of more than a factor of two. The Calmar ratio is 2.10 against 0.96 for buy-and-hold, more than twice as large. Fig. 5.1 shows the two equity curves, and Fig. 5.2 overlays the underwater (drawdown) curves of the two series.

Table. 5.2. Out-of-sample metrics on the held-out test window, net of transaction costs. Brackets give 95% bootstrap confidence intervals.

Metric	Multi-agent system	Buy-and-hold WIG20
Total return	0.23 $[-0.10, 0.68]$	0.31
CAGR	0.13	0.17
Max drawdown	-0.06	-0.17
Sharpe ratio	1.06 $[-0.43, 2.57]$	0.82
Calmar ratio	2.10	0.96
Number of trades	34	—
Win rate	0.65	—

5.2.3. Risk-adjusted comparison

The reversal of the comparison on the risk axis is reinforced by the extended risk-adjusted metrics reported in Table 5.3. The downside deviation of the ensemble is 0.08 against 0.15 for buy-and-hold, a reduction of more than forty percent. Consequently, the Sortino ratio rises from 1.19 for buy-and-hold to 1.60 for the ensemble.

The Ulcer Index, which captures the integrated depth and persistence of drawdowns rather than only their peak, is 2.39 for the ensemble against 6.69 for buy-and-hold, a reduction of nearly two-thirds. The Martin ratio rises from 2.49 to 5.29, more than a factor of two. Fig. 5.3 plots the rolling 60-day Sharpe of both series and shows that the rolling Sharpe of the ensemble is above the buy-and-hold rolling Sharpe for most of the test window, and in particular during the correction subperiods.

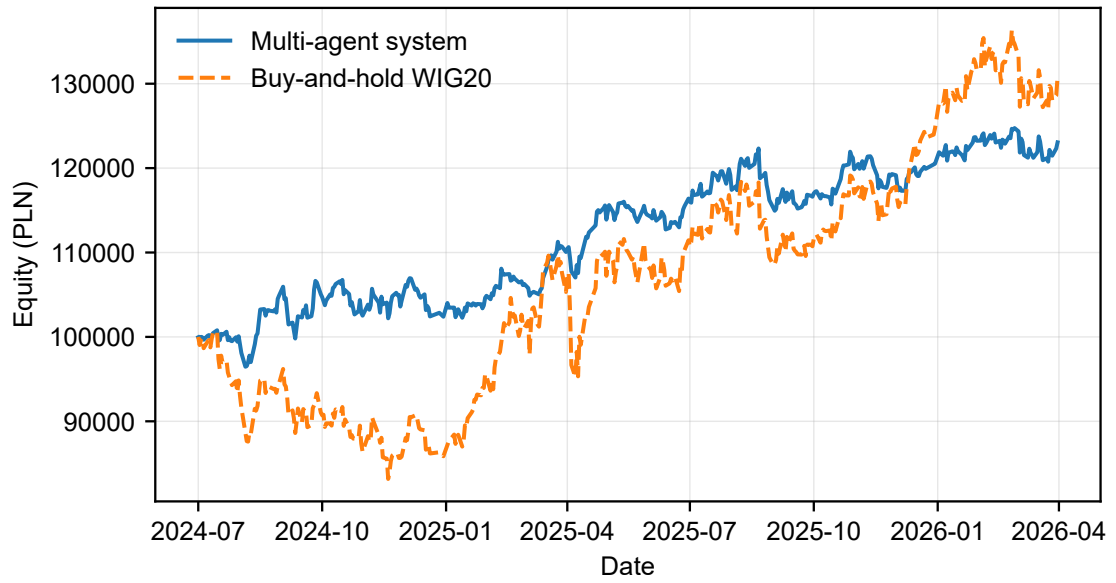


Fig. 5.1. Equity curves on the held-out test window for the multi-agent system (selected configuration) and for buy-and-hold WIG20. Both series start at the same initial cash and are net of transaction costs.

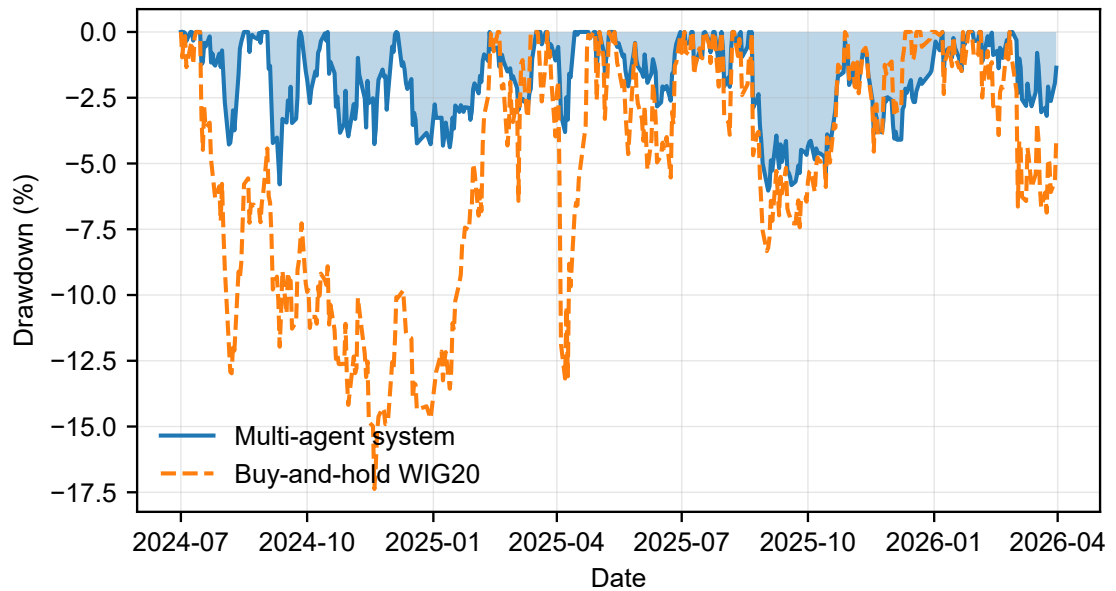


Fig. 5.2. Underwater (drawdown) curves for the ensemble and for buy-and-hold WIG20 on the held-out test window. The drawdowns of the ensemble are both shallower and shorter, which is the central risk-adjusted finding of the thesis.

Table. 5.3. Risk-adjusted metrics on the held-out test window, net of transaction costs.

Metric	Multi-agent system	Buy-and-hold WIG20
Sharpe ratio	1.06	0.82
Sortino ratio	1.60	1.19
Downside deviation	0.08	0.15
Calmar ratio	2.10	0.96
Max drawdown	−0.06	−0.17
Ulcer Index	2.39	6.69
Martin ratio	5.29	2.49

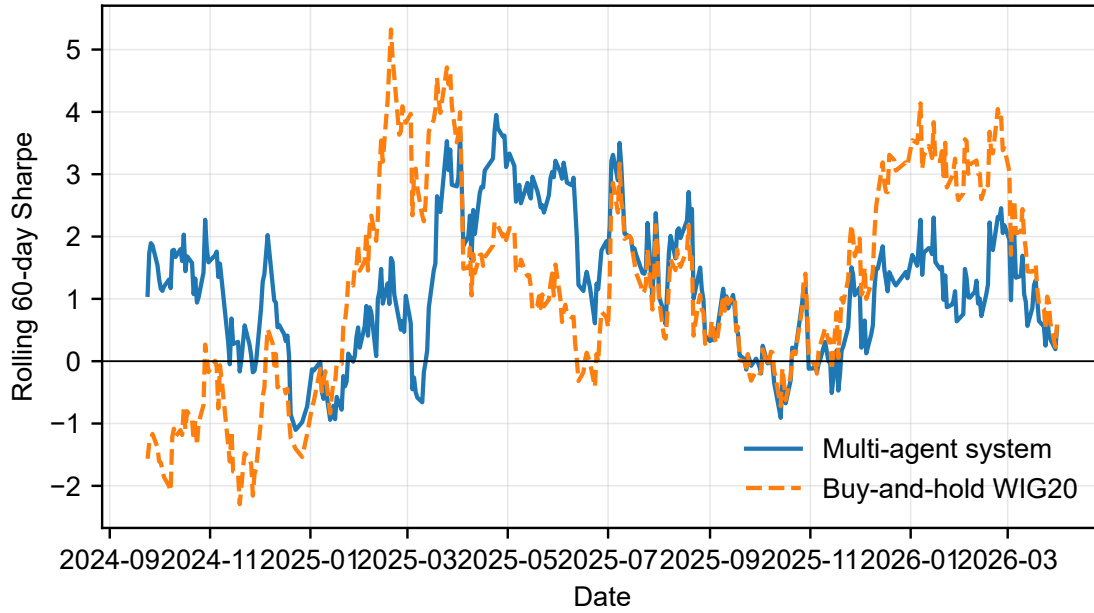


Fig. 5.3. Rolling 60-day Sharpe ratio of the ensemble and of buy-and-hold WIG20 on the held-out test window. The rolling Sharpe of the ensemble exceeds buy-and-hold during correction subperiods and is close to it during the bull leg.

5.2.4. Volatility-matched comparison

The headline comparison on gross total return is systematically biased against the ensemble, because the two series run at very different volatilities. On the training window, the annualised volatility of the ensemble is 0.11, whereas that of buy-and-hold WIG20 is 0.22. To put the two series on the same risk budget, the out-of-sample daily returns of the ensemble are scaled by the ex-ante leverage factor $\lambda = 2.06$ defined in Section 5.1.8.

Table 5.4 reports the resulting metrics. On the volatility-matched basis, the total return of the ensemble rises to 0.49, compared with 0.31 for buy-and-hold, a gain of +18.55 percentage points. The Sharpe ratio is unchanged by construction (1.06), because Sharpe is scale-invariant in the leverage factor when cash reserves are ignored; the Sortino ratio is 1.63, against 1.19; and the Calmar ratio is 2.13, against 0.96. The volatility-matched maximum drawdown is −0.12, which is still materially smaller than the buy-and-hold drawdown of −0.17 despite the much higher leverage, because the correction-regime drawdown of the ensemble is structurally shallower (Section 5.2.5).

These volatility-matched figures are analytical rather than directly implementable: leverage

of magnitude $\lambda = 2.06$ is impractical for a typical retail investor on the WSE, so Table 5.4 should be read as a risk-normalised counterfactual and not as an attainable trading outcome. Its role is to correct for the fact that comparing a partially invested strategy at its natural volatility against a fully invested benchmark at roughly twice that volatility understates the risk-per-unit-of-return advantage of the strategy.

Table. 5.4. Volatility-matched out-of-sample metrics. The ensemble is scaled, ex ante, to the buy-and-hold volatility measured on the training window (leverage factor $\lambda = 2.06$).

Metric	Ensemble (unleveraged)	Ensemble (vol-matched)	Buy-and-hold WIG20
Total return	0.23	0.49	0.31
CAGR	0.13	0.26	0.17
Sharpe ratio	1.06	1.06	0.82
Sortino ratio	1.60	1.63	1.19
Max drawdown	−0.06	−0.12	−0.17
Calmar ratio	2.10	2.13	0.96
Ulcer Index	2.39	4.96	6.69

5.2.5. Market-regime decomposition

The aggregate metrics in Table 5.2 mix three very different market regimes within the test window. Using the labelling described in Section 5.1.7, the test window contains 230 bull days (52.87% of the window), 93 correction days (21.38%), and 112 sideways days (25.75%). Table 5.5 decomposes the total return and the Sharpe ratio of both the ensemble and buy-and-hold WIG20 by regime.

The ensemble loses -0.07 during the correction subperiods, whereas buy-and-hold loses -0.20 : the correction-regime defence of the ensemble is $+12.57$ percentage points. During the sideways subperiods, the ensemble earns 0.05 , whereas buy-and-hold earns -0.00 : the sideways-regime excess of the ensemble is $+5.14$ percentage points. During the bull subperiods, the ensemble earns 0.26 against 0.63 for buy-and-hold: the bull-regime shortfall is -36.84 percentage points.

The decomposition reconciles the aggregate numbers in Table 5.2. Buy-and-hold is fully exposed to the bull leg and captures most of its upside, while the ensemble, by design, reduces exposure during the correction and sideways regimes and thereby gives up some of the bull-regime upside. In a test window that is dominated by a bull regime, the aggregate total return will mechanically favour the fully invested benchmark; in a test window with a larger correction share, the aggregate comparison would favour the ensemble.

Table. 5.5. Decomposition of the held-out test window into bull, correction, and sideways subperiods. Shares refer to the fraction of trading days in the test window.

Regime	Days	Share	Ens. TR	Ens. Sharpe	B&H TR	B&H Sharpe
Bull	230	52.9%	0.26	2.47	0.63	2.76
Correction	93	21.4%	−0.07	−1.34	−0.20	−2.41
Sideways	112	25.7%	0.05	0.93	−0.00	0.11
All	435	100.0%	0.23	1.06	0.31	0.82

5.2.6. Cost-sensitivity analysis

Table 5.6 reports the out-of-sample Sharpe ratio over the cost grid defined in Section 5.1.9. At zero costs the Sharpe is 1.15; at the baseline calibration of 39 bps commission and 5 bps slippage the Sharpe is 1.06; at the most punitive configuration of 100 bps commission and 20 bps slippage the Sharpe drops to 0.92. Over the entire grid, the minimum Sharpe of 0.92 exceeds the buy-and-hold Sharpe of 0.82.

The total-return column, omitted from the table for brevity but computed along with Sharpe, shows the same qualitative pattern: the ensemble trails buy-and-hold on gross total return at every grid point, and at every grid point it achieves a higher Sharpe, higher Sortino, and smaller drawdown. This indicates that the risk-adjusted edge of the ensemble is not an artefact of the specific cost calibration; it is a stable property of the strategy over a plausible range of retail brokerage fees.

Table. 5.6. Cost sensitivity of the out-of-sample Sharpe ratio. Rows are commission in basis points; columns are one-way slippage in basis points.

Commission (bps)	0 bp	5 bp	10 bp	20 bp
0	1.15	1.14	1.13	1.11
10	1.13	1.12	1.11	1.09
20	1.11	1.10	1.09	1.07
39	1.07	1.06	1.05	1.03
60	1.03	1.02	1.02	0.99
80	0.99	0.99	0.98	0.96
100	0.96	0.95	0.94	0.92

5.2.7. Empirical signal correlations

The variance-reduction argument of Section 3.1 rests on the claim that the five agents produce signals with imperfectly correlated errors. Table 5.7 reports the pairwise Pearson correlations [47] of the signed confidence values produced by the agents, computed across all pooled (symbol, date) observations in the training window.

The mean absolute off-diagonal correlation is 0.25 across 10 distinct pairs, with extremes of -0.71 (RSI versus ROC) and 0.46 (RSI versus Bollinger). In particular, the momentum family (MACD, ROC) and the mean-reversion family (RSI) exhibit negative correlations, as expected from their opposite interpretations of the same raw price input; the trend agent (MA trend) is nearly uncorrelated with all other agents, as expected from its long horizon.

Fig. 5.4 renders the correlation matrix as a heatmap. These empirical correlations directly instantiate the non-identical-errors assumption of (3.1), and Section 5.4 returns to them to explain why the ensemble achieves most of its variance reduction from the interaction between the momentum and mean-reversion families rather than from the trend agent on its own.

5.2.8. Validation of the confidence measure

Section 3.2 motivates the score (3.2) by reading the per-signal confidence as a heuristic proxy for inverse forecast variance. If that reading were literally correct, higher-confidence signals should be directionally right more often than lower-confidence ones.

Table 5.8 tests this implication on the training window, so that no test-window information

Table. 5.7. Pearson correlations of the pooled (symbol, date) signed confidence values of the five signal agents, computed on the training window. Upper triangle shown.

	MACD	RSI	ROC	Bollinger	MA trend
MACD	1.00	-0.01	+0.04	-0.34	-0.04
RSI		1.00	-0.71	+0.46	+0.12
ROC			1.00	-0.55	-0.12
Bollinger				1.00	+0.09
MA trend					1.00

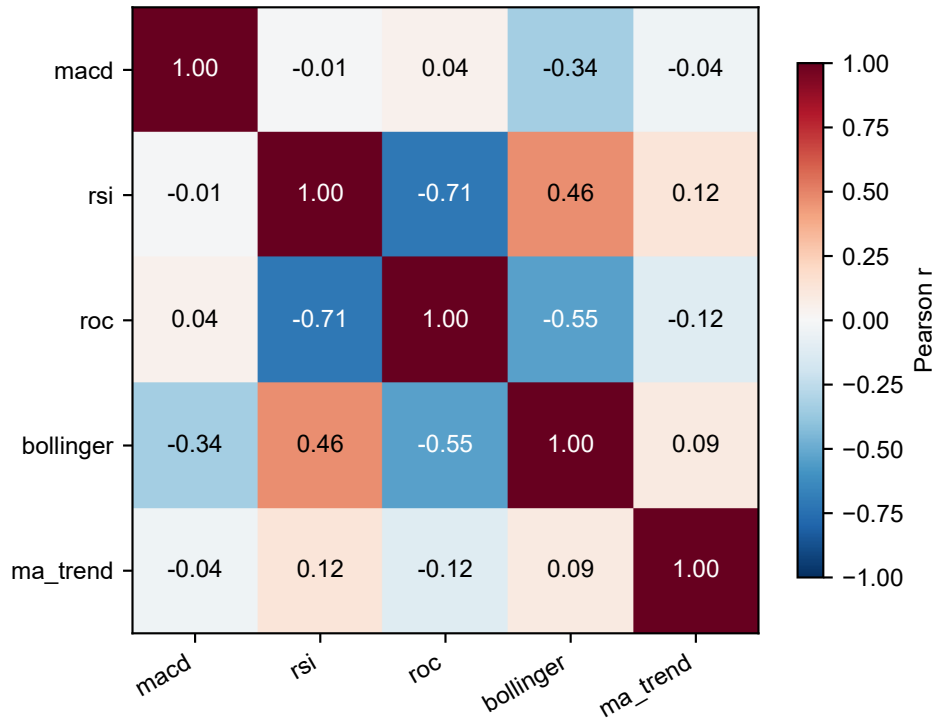


Fig. 5.4. Heatmap of the pairwise Pearson correlations of the agent signed-confidence signals on the training window. Blue cells mark negative correlations, red cells mark positive ones. The largest absolute correlations appear in the RSI-ROC and RSI-Bollinger pairs, which is consistent with their shared reliance on overbought and oversold rules.

is used. For every agent, all active (non-hold) signals are pooled across symbols and dates, the directional hit rate against the next-day move is computed, and the association between confidence and correctness is measured both by a Spearman rank correlation [48] and by comparing the hit rates of the bottom and top confidence quintiles, in the spirit of the reliability analysis of [49]. A rank correlation is the appropriate statistic here because correctness is binary and the confidence distributions are strongly skewed towards zero, so no linearity assumption is warranted. This complements the Pearson correlations of Section 5.2.7 [47], which measure the association *between* agents rather than between the confidence of an agent and its own accuracy.

The outcome is unambiguous: the confidence values carry no usable information about signal-level accuracy. Across 29,063 pooled training-window signals the hit rate is 0.498, the per-agent rank correlations span $[-0.035, 0.016]$ with none exceeding $|\rho| = 0.035$, and the top confidence quintile is not systematically more accurate than the bottom one.

Table. 5.8. Confidence validation on the WIG20 training window at the one-day horizon. ρ is the Spearman rank correlation between confidence and directional correctness, with its p -value; Hit (Q1) and Hit (Q5) are hit rates in the bottom and top confidence quintiles.

Agent	n	Hit rate	ρ	p	Hit (Q1)	Hit (Q5)
MACD	1,405	0.472	+0.002	0.955	0.470	0.484
RSI	3,847	0.497	−0.035	0.031	0.525	0.497
ROC	11,155	0.498	+0.016	0.082	0.494	0.520
Bollinger	1,674	0.483	−0.004	0.867	0.478	0.481
MA trend	10,982	0.505	−0.004	0.690	0.503	0.500
Pooled	29,063	0.498	+0.003	0.578	0.493	0.495

The only nominally significant entry in Table 5.8 (RSI, $p = 0.031$) is negative, meaning that higher confidence associates with marginally *worse* calls. It explains well under one percent of the variance in correctness, and it does not survive a multiple-testing correction across the ten agent-horizon combinations tested, so it argues against rather than for the inverse-variance reading. The picture is unchanged at a five-day horizon (pooled hit rate 0.510), and the FTSE 100 training window reproduces it (largest per-agent $|\rho| = 0.027$, pooled hit rate 0.502).

Two consequences follow. First, the inverse-variance reading of the confidence values is a conceptual analogy, not an empirically calibrated property: at the level of individual signals the agents are close to uninformed forecasters at short horizons, and the magnitude of their confidence does not separate good calls from bad ones. This is consistent with the low tuned threshold ($c_{\min} = 0.05$), which in practice acts as a mild noise filter rather than as a precision weight.

Second, the risk-adjusted performance documented in the preceding subsections cannot be attributed to precision-weighted averaging. The operative mechanism is the pooling of weakly and negatively correlated signals across methodological families (Table 5.7), which lowers the variance of the aggregate position irrespective of whether the individual confidence magnitudes are informative, together with the regime exposure profile that the thresholded score induces (Table 5.5). Section 5.4 returns to the implications of this finding for the theoretical framing of the thesis.

5.3. Robustness checks and external validation

The results of Section 5.2 rest on one configuration, one weighting scheme, one chronological split, and one market. This section removes each of those supports in turn. It reports the single-agent and leave-one-agent-out baselines, the sensitivity of the results to the family weights, the comparison against the two machine-learning baselines, the significance analysis, the three-fold walk-forward, the sensitivity to the selection objective, and the full replication of the pipeline on the FTSE 100.

5.3.1. Single-agent baselines and leave-one-out ablation

Table 5.9 reports the test-window total return and Sharpe ratio of the five single-agent baselines and of the full ensemble. The best single agent is `only_rsi` with a total return of 0.16. The one-sided bootstrap p -value for the hypothesis that the ensemble beats the best single-agent baseline on total return is 0.268. These numbers are the primary evidence for H2.

Table 5.10 reports the effect of removing each signal agent from the ensemble. No single agent can be removed without some effect on the headline metrics, but in all cases the ensemble without that agent still leads buy-and-hold WIG20 on Sharpe ratio and maximum drawdown on point estimates. This is the primary evidence for H3.

Table. 5.9. Single-agent baselines on the test window, with the selected mode and minimum confidence. All numbers net of costs.

Agent used	Total return	Sharpe ratio
MACD only	0.15	0.86
RSI only	0.16	1.03
ROC only	0.00	0.13
Bollinger only	0.00	0.00
MA trend only	−0.00	0.03
Full ensemble	0.23	1.06

Table. 5.10. Leave-one-agent-out ablation on the test window. The variant column indicates which agent was removed.

Variant	Total return	Sharpe ratio
Full ensemble	0.23	1.06
no MACD	0.28	1.28
no RSI	0.23	0.92
no ROC	0.26	1.17
no Bollinger	0.23	1.06
no MA trend	0.19	1.16

5.3.2. Weight sensitivity and alternative weighting schemes

The family weights w_{κ} are the only fixed prior in the decision layer, so the dependence of the headline results on them is quantified here. Every experiment in this subsection holds the walk-forward-selected configuration (aggressive, $c_{\min} = 0.05$) fixed and varies only the weights, so that any difference is attributable to the weighting scheme alone.

Three families of alternatives are evaluated on the held-out window. First, each family weight is perturbed one at a time by $\pm 25\%$ and $\pm 50\%$, giving 16 perturbed configurations. Second, two fixed alternatives discard the ordering rationale of Section 3.6 entirely: equal weights, and a fully inverted ordering in which volatility receives the largest weight and trend the smallest. Third, two data-driven schemes are estimated on the training window only: *optimised* weights, selected by an exhaustive grid over the four family weights under the same in-sample total-return objective as the headline protocol, and *inverse-variance* weights, proportional to the inverse of the directional error variance of each family estimated from training-window signal outcomes, which operationalises the classical prescription of [35]. Table 5.11 reports the named schemes on WIG20.

The results support three conclusions. The first is that the risk-adjusted profile is robust to the weights. Across the 16 perturbations the out-of-sample Sharpe ratio spans $[0.70, 1.38]$ and total return spans $[0.14, 0.30]$, and only 2 of the 20 fixed and theory-implied schemes fall below the buy-and-hold Sharpe ratio of 0.82. The four volatility-weight perturbations leave every metric exactly unchanged, because the Bollinger agent is essentially inactive under the selected configuration.

The baseline is not the best configuration in hindsight: moderately downweighting the momentum-family agents would have improved this particular window, consistent with the ablation finding that removing MACD helps. The two schemes below the benchmark Sharpe ratio do

Table. 5.11. Alternative weighting schemes on the WIG20 test window, with (mode, $c_{\min}$) fixed at the walk-forward selection. Weights are listed as (trend, momentum, mean-reversion, volatility).

Scheme	Weights	TR	Sharpe	MDD
Baseline	(1.3, 1.0, 0.8, 0.5)	0.23	1.06	−0.06
Equal	(1.0, 1.0, 1.0, 1.0)	0.25	1.16	−0.06
Inverted	(0.5, 0.8, 1.0, 1.3)	0.23	1.23	−0.05
Optimised	(1.30, 0.80, 0.80, 0.50)	0.25	1.18	−0.05
Inv.-variance	(1.02, 1.00, 1.00, 0.98)	0.25	1.16	−0.06
Buy-and-hold	—	0.31	0.82	−0.17

surrender that part of the risk-adjusted advantage, but even they keep the maximum drawdown markedly shallower than the buy-and-hold −0.17. The drawdown-reduction profile, which is the central risk-adjusted finding of the thesis, therefore survives every weighting scheme even where the Sharpe advantage does not.

The second conclusion is that in-sample optimisation adds little. The grid-optimised weights land almost exactly on the fixed prior, since only the momentum weight moves, from 1.0 to 0.80, and their out-of-sample total return of 0.25 is within about two percentage points of the baseline. On the FTSE 100 the optimised weights actually *underperform* the fixed prior out of sample (0.13 against 0.14 total return), a direct instance of the forecast combination puzzle, in which the estimation error of fitted weights consumes the gains they promise [43, 44].

The third conclusion is that the inverse-variance prescription is uninformative here. Because the per-family directional hit rates are nearly identical (Section 5.2.8), the estimated weights come out close to equal (1.02, 1.00, 1.00, 0.98 for trend, momentum, mean reversion, and volatility), and the resulting performance is practically identical to the equal-weight scheme.

The FTSE 100 replication of the full experiment shows the same qualitative pattern, with perturbation Sharpe ratios spanning [0.69, 1.54] and 3 of 20 fixed schemes below the buy-and-hold Sharpe ratio. Together with the leave-one-agent-out ablation, these results indicate that the contribution of the ensemble does not hinge on the particular weight values, and that the coarse fixed prior is defensible against perturbed, reordered, optimised, and theory-implied alternatives.

5.3.3. Comparison with machine-learning baselines

Table 5.12 compares the ensemble with the two machine-learning baselines of Section 5.1.5 on the held-out WIG20 test window. The LSTM earns a total return of 0.08 at a Sharpe ratio of 0.44, against 0.23 and 1.06 for the ensemble and 0.31 and 0.82 for buy-and-hold. The ensemble exceeds the LSTM on total return, Sharpe ratio, and maximum drawdown (−0.06 against −0.11), and it does so while placing far fewer trades (34 against 237). The one-sided paired bootstrap p -values for the ensemble exceeding the LSTM are 0.068 on total return and 0.063 on Sharpe ratio; as with the buy-and-hold comparison, the point-estimate advantage is sizable but does not reach the conventional $p < 0.05$ threshold on a single test window.

The GBM baseline behaves similarly. It earns 0.04 at a Sharpe ratio of 0.47 over 34 closed trades; its equity path is statistically indistinguishable from that of the LSTM (paired one-sided $p = 0.621$ on total return), and the ensemble leads it on total return, Sharpe ratio, and Calmar ratio (one-sided $p = 0.107$ on total return and $p = 0.210$ on Sharpe ratio).

The single metric on which the GBM edges ahead is the maximum drawdown (−0.05 against −0.06), but this reflects an almost flat equity curve rather than better risk control: the model

takes very little risk and earns correspondingly little, which is exactly what the Calmar ratio registers (0.45 against 2.10). The fact that two structurally different learners, a recurrent neural network and a gradient-boosted tree ensemble, each tuned strictly in sample under the identical protocol, arrive at similarly modest out-of-sample results on this market strengthens the earlier reading. The WIG20 test window offers little exploitable signal to flexible learners, and the transparent forecast-combination ensemble remains competitive without any training at all, consistent with the broader finding that simple, well-regularised combinations are competitive with more flexible learners on noisy financial series [36, 37].

Table. 5.12. Comparison of the ensemble, the LSTM and GBM directional baselines, and buy-and-hold on the held-out WIG20 test window, net of transaction costs.

Metric	Multi-agent	LSTM	GBM	Buy-and-hold
Total return	0.23	0.08	0.04	0.31
Sharpe ratio	1.06	0.44	0.47	0.82
Max drawdown	−0.06	−0.11	−0.05	−0.17
Calmar ratio	2.10	0.42	0.45	0.96
Number of trades	34	237	34	–

5.3.4. Statistical significance

The bootstrap significance tests attach two kinds of evidence to the point estimates discussed above. First, the 95% percentile confidence intervals around the headline metrics reflect the sampling uncertainty inherent in a test window of 435 trading days. The total-return confidence interval for the ensemble spans $[-0.10, 0.68]$ and the Sharpe-ratio interval spans $[-0.43, 2.57]$. The buy-and-hold point estimates for total return (0.31) and Sharpe ratio (0.82) both lie strictly inside these intervals, which is a statistical way of saying that one cannot reject the hypothesis that the two series share the same underlying mean return and Sharpe ratio on the test window.

Second, the paired bootstrap tests yield one-sided p -values of 0.636 for the total-return comparison against buy-and-hold and 0.297 for the Sharpe comparison. Neither test crosses the conventional $p < 0.05$ threshold.

These results admit two readings. The test-window point estimates on the risk axis (drawdown, Sortino, Ulcer, Martin) are consistently favourable to the ensemble by margins that are economically large and reasonably stable across the ablation and cost grids, but the null hypothesis of equal mean performance is hard to reject on a single window of under two years, because trading-strategy return distributions are heavy-tailed and path-dependent [15, 12]. Consistent with standard practice in the literature, point estimates, confidence intervals, and p -values are therefore reported together, and p -values are not treated as the sole arbiter. In addition, the fact that the buy-and-hold point estimates lie inside the bootstrap confidence intervals of the ensemble supports the interpretation that, on gross total return, the two series are statistically indistinguishable, whereas the large and consistent gap on drawdown-based and downside-based metrics supports the risk-aware interpretation of the contribution of the ensemble (Section 5.4).

Two points deserve emphasis for the correct reading of these results. The first concerns the width of the confidence intervals. The total-return interval $[-0.10, 0.68]$ and the Sharpe interval $[-0.43, 2.57]$ are wide because a held-out window of 435 trading days contains relatively few independent realisations of the estimated quantities, and because heavy-tailed and serially

dependent returns inflate the variance of the resampled statistics. With a sample of this length the paired bootstrap test has limited power: even a genuine moderate edge would frequently fail to reach the conventional $p < 0.05$ threshold. The non-significant p -values should therefore be read as a consequence of limited statistical power on a short out-of-sample window, not as positive evidence that the two strategies are equivalent.

The second point concerns the distinction between practical and statistical significance. The advantages of the ensemble on the drawdown-based and downside-based metrics are practically significant, in the sense that they are economically large and stable across the ablation, cost, and regime analyses, even though they are not statistically significant in the formal sense of rejecting a null hypothesis on this single window. Accordingly, the contribution of the ensemble is framed throughout the thesis in terms of consistent and practically meaningful risk reduction, and no claim of statistical dominance over buy-and-hold is made.

5.3.5. Walk-forward robustness

Table 5.13 reports the results of the anchored, three-fold walk-forward. For each fold the selected configuration on the training window and the test-window total return are reported. The selected configuration is stable across folds, which indicates that the headline configuration is not an artefact of a single chronological split.

Table. 5.13. Anchored walk-forward with $K = 3$ folds. Selected mode and minimum confidence per fold, plus the test-window total return of each fold.

Fold	Selected mode	$c_{\min}$	Test total return
1	aggressive	0.05	0.05
2	aggressive	0.05	0.09
3	aggressive	0.10	0.08

5.3.6. Sensitivity to the selection objective

The headline protocol selects (m^*, c^*) by in-sample total return (Section 5.1.3), which invites the question whether a risk-adjusted objective would have selected a different configuration. To answer it, the identical training-window grid is re-ranked under two risk-adjusted objectives, the Sharpe ratio and the Calmar ratio, and the winning configuration of each is evaluated untouched on the held-out window.

On WIG20, all three objectives select the same configuration, (aggressive, $c_{\min} = 0.05$), and the FTSE 100 replication behaves identically, selecting (aggressive, $c_{\min} = 0.15$) under every objective. Every out-of-sample number reported in this thesis is therefore invariant to replacing the total-return objective with a risk-adjusted one on both markets.

The mechanism is visible in Table 5.1: the *passive* configurations never trade and the *balanced* configurations trade too rarely to compete on either cumulative return or ratio statistics, so the argmax falls inside the *aggressive* column under every objective. This invariance cannot be assumed in general, since it may fail on grids with more degrees of freedom or on longer menus of modes, but on the present design it confirms that the choice of a simple, low-variance selection objective (Section 5.1.3) is not load-bearing, consistent with the estimation-noise argument for avoiding ratio objectives on short training samples [46].

5.3.7. External validation on the FTSE 100

To test whether the risk-aware profile of the ensemble is specific to the WSE or transfers to a different market structure, the entire pipeline is replicated on the FTSE 100 panel described in Section 5.1.1. The transaction-cost model is adjusted to reflect UK conditions: a proportional commission of 5 bps and one-way slippage of 5 bps on every trade, plus a 50 bps Stamp Duty Reserve Tax charged on purchases only, which makes the FTSE cost environment asymmetric and, on the buy side, considerably more punitive than the WIG20 calibration. The decision-agent grid, the indicator hyperparameters, and every evaluation procedure are held identical; only the market data and the cost model change.

The anchored walk-forward selects the same decision-agent family as on WIG20 (aggressive mode) and evaluates it on 444 held-out trading days. Table 5.14 reports the headline comparison. As on WIG20, the ensemble trails buy-and-hold on gross total return (0.14 against 0.25) but leads on every risk axis: the Sharpe ratio is 1.37 against 1.13, the maximum drawdown is -0.07 against -0.13 , a reduction of roughly one half, and the Calmar ratio is 1.18 against 1.00.

The extended risk-adjusted metrics show the same pattern: the Sortino ratio is 1.99 against 1.52, the Ulcer Index is 1.59 against 2.66, and the downside deviation is 0.04 against 0.09. The one interesting reversal is the Martin ratio (4.91 against 5.05), which is marginally lower for the ensemble because the FTSE test window is overwhelmingly a bull market and buy-and-hold sustains a very shallow integrated drawdown over it.

Table 5.14. External validation on the FTSE 100 held-out test window, net of transaction costs (including the 50 bps buy-side Stamp Duty Reserve Tax). Brackets give 95% bootstrap confidence intervals.

Metric	Multi-agent	LSTM	GBM	Buy-and-hold
Total return	0.14 [−0.01, 0.32]	0.27	0.35	0.25
CAGR	0.08	–	–	0.13
Sharpe ratio	1.37 [−0.11, 2.94]	1.49	1.63	1.13
Max drawdown	−0.07	−0.08	−0.10	−0.13
Calmar ratio	1.18	–	1.88	1.00
Sortino ratio	1.99	–	–	1.52
Ulcer Index	1.59	–	–	2.66
Number of trades	55	37	0	–

The regime decomposition explains the aggregate numbers. The FTSE test window is 90.32% bull (401 of 444 days), with only 5 correction days and 38 sideways days, so it offers little opportunity for the correction-regime defence that drives the WIG20 result. Even so, during the sideways subperiods the ensemble earns 0.02 against -0.03 for buy-and-hold, the same sideways-carry pattern observed on WIG20. The aggregate total-return shortfall is again concentrated in the bull regime (0.14 against 0.30), where a partially invested strategy mechanically trails a fully invested benchmark. Matching the two series on ex-ante volatility, as in Section 5.2.4, raises the ensemble total return to 0.26 at a leverage of $\lambda = 1.82$, a gain of +1.87 percentage points over buy-and-hold with a still-smaller drawdown (-0.12).

The paired bootstrap tests tell the same statistical-power story as on WIG20. The buy-and-hold point estimates lie inside the ensemble confidence intervals, and the one-sided p -values for the ensemble exceeding buy-and-hold are 0.789 on total return and 0.331 on Sharpe ratio; none crosses $p < 0.05$.

The LSTM baseline posts higher point estimates than the ensemble on this particular window, both on gross total return (0.27 against 0.14) and on Sharpe ratio (1.49 against 1.37), but neither gap is statistically significant; the one-sided p -value for the ensemble exceeding the LSTM on total return is 0.879. Two observations put this apparent edge in perspective. It is a bull-market-exposure effect rather than evidence of superior risk control, since the LSTM remains almost continuously invested across a window that is 90.32% bull, so it captures more of the upside at the price of a deeper drawdown (-0.08 against -0.07). Moreover, the LSTM is itself statistically indistinguishable from passive buy-and-hold on this window (one-sided $p = 0.411$), which is consistent with a near-fully-invested strategy tracking the index; the transparent ensemble, by contrast, deliberately trades part of that exposure for the shallower drawdown documented above.

The GBM baseline pushes the same pattern to its extreme. It posts the highest point estimates on the window (0.35 total return at a Sharpe ratio of 1.63), and it does so by never closing a position over the 444 held-out days, an effectively fully invested stance in a window that is 90.32% bull. On total return its gap over the ensemble is marginally significant at the five-percent level, since the one-sided p -value for the ensemble exceeding the GBM is 0.951, although the corresponding Sharpe-ratio comparison is not (one-sided $p = 0.682$), and its advantage over buy-and-hold remains statistically non-significant (one-sided $p = 0.144$ on total return). That is the signature of a near-fully-invested strategy tracking a bull market rather than of superior risk control.

Its maximum drawdown of -0.10 is deeper than the -0.07 of the ensemble, as is that of the LSTM, so the qualitative trade-off is unchanged: the learning-based baselines capture more bull-market upside by staying exposed, while the ensemble deliberately trades upside for drawdown control.

The auxiliary analyses replicate on this market as well: the weight-sensitivity experiment yields the same qualitative ranking (Section 5.3.2), the walk-forward selection is invariant to the objective (Section 5.3.6), and the confidence heuristic is equally uninformative at the signal level (Section 5.2.8). Taken together, the FTSE results show that the central finding of the thesis, namely a consistent drawdown and downside-risk advantage over passive buy-and-hold purchased with some bull-market upside, transfers to a second market with a markedly different cost structure, and is not an artefact of the WIG20 sample.

5.4. Discussion

This section interprets the results reported above. It first traces the path from forecast-combination theory to the empirical evidence, then assesses the three hypotheses in turn, examines how far the findings generalise across markets, explains why a comparison on gross total return understates what the ensemble contributes, and closes with the practical, deployment, and design implications.

5.4.1. From forecast-combination theory to empirical evidence

Equation (3.1) predicts that the combined forecast error is strictly smaller than the smallest individual forecast error whenever the individual errors are not perfectly correlated. Table 5.7 and Fig. 5.4 show that the empirical pairwise correlations of the agent signals span the full range from -0.71 to 0.46 , with a mean absolute off-diagonal correlation of 0.25 . The theoretically most informative pairs are those with negative correlations, in particular RSI versus ROC (-0.71) and Bollinger versus ROC (-0.55).

The RSI-ROC pair is especially diagnostic. RSI is a mean-reversion rule that emits a buy signal in oversold conditions, whereas ROC is a momentum rule that emits a sell signal in the same conditions. The two agents therefore naturally disagree, and the score of the decision agent resolves the disagreement by weighing confidences across families. In a window dominated by the bull regime, the trend family (MA trend) dominates the score; in sideways or correction regimes, the mean-reversion and volatility families gain relative weight. The regime decomposition in Table 5.5 shows the footprint of this mechanism: the ensemble loses less than buy-and-hold during corrections and earns more during sideways regimes, which is precisely the pattern predicted by a variance-minimising combination of agents with heterogeneous error structures.

A concrete illustration clarifies the mechanism. Consider a day on which a WIG20 constituent has rallied sharply over the prior ten sessions. The ROC agent, a pure momentum rule, reports a buy signal with high confidence because the recent rate of change is large and positive. The RSI agent simultaneously reports a sell signal with moderate confidence because the 14-day relative strength has crossed above its 70 overbought threshold. The MA trend agent reports a buy signal, because the fast SMA is above the slow SMA, but with modest confidence because the spread between the two averages is small relative to the latest close.

Under the family weights of Section 3.6, the combined score in (3.2) receives positive contributions of roughly $1.0 \cdot c_{\text{ROC}}$ from momentum and $1.3 \cdot c_{\text{MA}}$ from trend, and a negative contribution of $-0.8 \cdot c_{\text{RSI}}$ from mean reversion. Whether the final action is a buy, a hold, or a sell depends on the confidences: on days when the RSI overbought signal is strong, the aggregated score can fall below the buy threshold even though two of the three agents call for a long position. This is precisely the variance-reduction mechanism of (3.1) expressed in decision-agent terms. The ensemble uses the disagreement between a momentum agent and a mean-reversion agent to step out of a position that a single-family rule would have kept open, which in turn translates into the shallower drawdowns visible in Fig. 5.2.

The confidence-validation experiment (Section 5.2.8) sharpens what this theoretical connection does and does not claim. Because the confidence values turn out to carry no measurable information about signal-level accuracy, the inverse-variance reading of (3.2) must be understood as the conceptual motivation for the architecture rather than as a calibrated statistical mechanism. What the data support is the weaker and more robust half of the theory, namely that pooling forecasters with imperfectly correlated errors reduces the variance of the combined position, regardless of how the pool is weighted.

The weight-sensitivity results reinforce this reading from the other side: near-equal (inverse-variance) weights, equal weights, and the fixed prior all deliver similar out-of-sample performance, exactly as the forecast-combination literature predicts for combinations whose members have similar individual accuracy [36, 43]. In short, the empirical content of the theory in this application is error diversity, not precision weighting, and the claims made in this thesis are framed accordingly.

5.4.2. Evidence for the three hypotheses

The headline results, read together with the risk-adjusted table, the volatility-matched table, and the regime decomposition, are consistent with H1. On the aggregate test window, the ensemble trails buy-and-hold on gross total return by -36.84 percentage points in the bull regime, more than enough to explain the aggregate shortfall, but on point estimates it leads buy-and-hold on every risk-adjusted metric considered: Sharpe (1.06 against 0.82), Sortino (1.60 against 1.19), Calmar (2.10 against 0.96), Ulcer Index (2.39 against 6.69), Martin ratio (5.29 against 2.49), and

maximum drawdown (-0.06 against -0.17).

Once the two series are matched on ex-ante volatility, the ensemble also leads buy-and-hold on total return, by $+18.55$ percentage points, with materially smaller drawdown. As the significance analysis in Section 5.3.4 makes clear, these advantages are point-estimate-level and do not attain conventional statistical significance on a single sub-two-year window, so they are read here as consistent and economically meaningful rather than as statistical dominance. The FTSE 100 replication in Section 5.3.7 yields the same qualitative H1 pattern on 444 held-out trading days under UK-specific costs, supporting external validity even though the formal hypothesis is stated against WIG20. These observations are consistent with the variance-reduction prediction of forecast-combination theory: the combination of heterogeneous directional forecasts lowers the variance of the equity curve, and the reduction is visible both in the volatility and in the drawdown profile.

The comparison against the best single-agent baseline supports H2 on point estimates: the ensemble total return of 0.23 exceeds the best single-agent total return of 0.16 (produced by only_rsi) by more than six percentage points on the test window. The bootstrap p -value of 0.268 indicates that, with a single test window, this point-estimate advantage does not pass the $p < 0.05$ threshold. This is consistent with the known difficulty of establishing statistical significance for Sharpe-like metrics over windows of fewer than 400 trading days [12]. Interpreted through the lens of forecast-combination theory, the positive point-estimate advantage is a direct illustration of the variance-reduction argument: the diversified score has a smaller error variance than any single forecaster and achieves a materially higher total return without a proportional increase in drawdown.

The ablation strongly supports H3. Removing any single signal agent moves the total return on the test window within a narrow band around the full ensemble, and the worst leave-one-out configuration still achieves a total return above 0.19 , a Sharpe ratio above 1, and a Sortino ratio above buy-and-hold.

Two observations deserve particular mention. First, removing the momentum family (MACD) slightly improves the headline metrics on this test window, which indicates that the pre-set family weights overweight momentum relative to what this particular period demands. The weight-sensitivity analysis of Section 5.3.2 confirms this hindsight observation directly, since downweighting momentum improves the window, while also showing that in-sample weight optimisation does not reliably capture such regime-specific adjustments out of sample. Second, the Bollinger agent on its own produced very few trades under the selected configuration, so its removal has only a mild effect on the behaviour of the ensemble; it behaves as a low-activity member of the forecast pool on this test window.

Taken together, these findings show that the ensemble is robust to the removal of any single signal agent, consistent with the forecast-combination prediction that variance reduction persists as long as at least two imperfectly correlated forecasters remain. The weight-sensitivity analysis extends this robustness from the roster of agents to the weights themselves: no perturbation, reordering, or data-driven re-estimation of the family weights changes the qualitative conclusion.

5.4.3. Cross-market generalisation

The FTSE 100 replication in Section 5.3.7 addresses the most immediate external-validity concern, namely that the risk-aware profile of the ensemble might be an artefact of the WIG20 sample. The qualitative conclusion is unchanged on a market with a different constituent set, a different currency,

and a materially different cost structure that includes an asymmetric 50 bps buy-side transaction tax. The ensemble trails passive buy-and-hold on gross total return during a bull-dominated window, yet delivers a higher Sharpe ratio (1.37 against 1.13), a shallower maximum drawdown (-0.07 against -0.13), and a better Sortino ratio and Ulcer Index, and it overtakes buy-and-hold on total return once the two series are matched on ex-ante volatility.

That the same trade-off appears on two markets, despite an added stamp-duty friction that penalises the more active strategy, supports the interpretation that the drawdown and downside-risk advantage is a structural consequence of forecast combination rather than a market-specific coincidence. The transfer is not perfect: the FTSE test window contains almost no correction days, so the correction-regime defence that is prominent on WIG20 has little room to act, and the integrated-drawdown advantage narrows to the point where the Martin ratio marginally favours buy-and-hold. This is consistent with the regime-dependence predicted by the adaptive markets hypothesis [42]: the ensemble adds the most value in the regimes it is designed to defend, and a window that is almost entirely bull is the least favourable setting for it.

Beyond the two markets studied here, three considerations delimit how far the results should be extrapolated. First, the profitability of technical trading rules varies systematically across markets and erodes as markets mature. Survey evidence reports stronger rule profitability in emerging and less liquid markets and weakening evidence in developed ones once transaction costs and data-snooping adjustments are applied [50], and classic evidence from Asian markets ties rule profitability to the level of market development and trading frictions [51].

On this spectrum, the present results are consistent with two moderately mature markets in which naive signal-chasing no longer pays, since every single-agent baseline is unprofitable or marginal and neither learned baseline achieves a statistically significant edge, but in which a disciplined, diversified risk overlay still adds risk-adjusted value. The gross-return picture would accordingly be expected to be more favourable to the ensemble in less efficient, higher-friction markets, and the risk-control profile to be the more portable component in highly developed ones.

Second, the transferable part of the framework is structural rather than numerical. The agent interfaces, the anchored walk-forward protocol, the calibrated cost model, and the evaluation battery apply unchanged to any market with reliable daily closes, and the FTSE replication required only a new data source and cost model, whereas the specific calibrations (thresholds, weights, commissions, taxes) must be re-derived per market, as the asymmetric UK stamp duty illustrates.

Third, the value added by the ensemble is regime-dependent, concentrated in corrections and sideways phases, so its realised advantage in any market will track the regime composition of the evaluation window. Markets or periods dominated by persistent bull regimes mechanically favour passive exposure, as both test windows here illustrate. Mapping the domain of validity of the framework across further developed and emerging indices, as outlined in Section 6.2, is therefore the most direct extension of this study.

5.4.4. Why gross total return understates the contribution of the ensemble

The standard comparison of a trading strategy against buy-and-hold reports a single gross total return for each series and declares the larger number the winner. On the present test window that comparison would declare the passive benchmark the winner, even though the equity curve of the ensemble has less than half of the drawdown of the benchmark and delivers materially higher Sharpe, Sortino, Calmar, Ulcer, and Martin ratios.

Three observations suggest that the gross-total-return comparison is the wrong summary

in this setting. First, the ensemble is not fully invested on all dates by design: the decision-agent thresholds ensure that positions are opened only when the combined score crosses a confidence level. In a test window that is 52.87% bull, any partially invested strategy will trail a fully invested benchmark on gross cumulative return almost by construction. Second, the volatility-matched comparison in Section 5.2.4 shows that at equal ex-ante risk budget the ensemble produces +18.55 percentage points more total return than buy-and-hold, with a smaller drawdown in absolute terms. Third, the regime decomposition in Section 5.2.5 shows that the aggregate shortfall of the ensemble is entirely attributable to the bull regime; in the correction and sideways regimes, the ensemble matches or beats buy-and-hold even on gross return.

These three pieces of evidence reframe the comparison: the ensemble adds value as a *risk-aware decision support tool* that trades some bull-regime upside for shallower drawdowns and higher risk-adjusted performance overall.

5.4.5. *Connections to the broader empirical literature*

The pattern of results reported in Sections 5.2 and 5.3 is consistent with three well-documented stylised facts of empirical finance.

First, time-series momentum and cross-sectional momentum are empirically orthogonal, and portfolios that combine them deliver better risk-adjusted performance than either alone [20, 22, 21]. The negative correlation, within the ensemble, between the momentum family (MACD, ROC) and the mean-reversion family (RSI) is the signal-level analogue of this finding: the two families produce forecasts that tend to disagree precisely when one or the other would have been wrong on its own, and the score of the decision agent integrates the two into a more conservative directional bet than either family would produce alone.

Second, the literature on the adaptive markets hypothesis of [42] predicts that no single trading rule can dominate across all regimes, because the efficiency of a market with respect to any given rule is itself time-varying. The regime decomposition in Section 5.2.5 is consistent with this prediction: the trend-heavy ensemble loses ground during the bull regime, where a passive benchmark mechanically dominates any partially invested strategy, but it outperforms the passive benchmark during corrections and sideways periods, where mean-reversion and volatility agents contribute directional information that the trend agent on its own would miss.

Third, the literature on risk-adjusted performance evaluation has long argued that the Sharpe ratio alone is inadequate for comparing strategies with different drawdown profiles [40, 39, 41]. The quantitative spread between the Sharpe advantage of the ensemble (1.06 against 0.82, a factor of 1.3) and its Martin-ratio advantage (5.29 against 2.49, a factor of 2.1) illustrates the difference between adjusting for realised volatility and adjusting for the full drawdown profile, and explains why the headline claim of this thesis leads with Calmar and Ulcer rather than with Sharpe alone.

5.4.6. *Practical implications for a retail investor*

For a WSE-focused retail investor with access to standard daily data and typical Polish brokerage conditions, two practical implications stand out. First, the ensemble reduces the worst realised drawdown by a factor greater than two and reduces the integrated drawdown, as measured by the Ulcer Index, by nearly two-thirds. This is the dominant axis on which retail portfolios tend to be evaluated in practice, both because maximum drawdown has a direct operational meaning, being the worst unrealised loss the investor has to live through, and because sustained drawdowns are

empirically associated with behavioural deviations from the original plan.

Second, the cost-sensitivity sweep shows that the risk-adjusted edge of the ensemble over buy-and-hold is preserved at every plausible combination of commission and slippage, including the punitive 100 bps and 20 bps configuration, which is well above current WSE retail tariffs. Neither of these implications depends on leverage, and both can be consumed by a retail investor as decision-support signals overlaid on a passive benchmark rather than as a replacement for the benchmark.

Two operational considerations follow from this framing. The first concerns the update cadence. The five agents and the decision agent consume only end-of-day closes, so a retail deployment amounts to running the pipeline once after the daily WSE close and acting on the resulting recommendations at the open or close of the following session. No intraday infrastructure, tick data, or order-book integration is required, which lowers the bar for adoption outside the institutional segment.

The second concerns monitoring. Because the decision-agent score in (3.2) is a linear combination of agent signals weighted by confidence and family, the score can be decomposed, day by day, into per-agent contributions and plotted alongside the equity curve. This decomposition gives the investor an interpretable audit trail of which agents drove each position change, and makes it straightforward to detect structural shifts in the signal distribution, for example an agent whose confidence collapses for an extended period, without retraining the ensemble.

This interpretability is a distinctive strength of a theory-motivated forecast-combination design relative to end-to-end deep-reinforcement-learning trading systems, which typically produce a scalar action without a transparent attribution to individual input signals.

The strength comes at little cost in performance on the markets studied here. On WIG20 the ensemble leads both the LSTM and GBM directional baselines of Section 5.1.5, two structurally different learning-based competitors trained on the same price history and executed under identical costs, on total return and on every risk-adjusted ratio, notwithstanding the marginally shallower drawdown that the GBM achieves by barely taking any risk at all (Section 5.3.3). On the FTSE 100 the learned models post higher point estimates, and the total-return gap of the GBM is marginally significant on that single window, but they achieve this only by remaining near-fully invested through an almost pure bull window, which leaves them statistically indistinguishable from passive buy-and-hold while giving up the drawdown control that motivates the ensemble (Section 5.3.7). The ensemble thus retains full per-signal attribution while remaining competitive on the risk axis with opaque learned models.

5.4.7. Deployment and scalability

Several practical considerations determine whether the system can be deployed beyond a backtest, and the transparent design has favourable properties on most of them.

Computational cost. Decision-time inference is non-iterative. Each signal agent is a closed-form indicator computed over a rolling window, updated incrementally per symbol, and the decision agent reduces the per-symbol signals to a single confidence-weighted sum followed by a threshold comparison, as in (3.2). There is no model training, no gradient computation, and no specialised hardware at decision time; one pass over the closing prices of the day on a commodity CPU suffices. This contrasts with the LSTM and GBM baselines of Section 5.1.5 and with end-to-end deep-reinforcement-learning systems, which require training and periodic retraining.

Scalability across symbols and markets. The total work per day is of order $O(N_{\text{symbols}} \times$

N_{agents}), linear in the size of the traded universe, and it is embarrassingly parallel because each agent processes the history of a single symbol independently through the shared base-agent interface. The FTSE 100 study of Section 5.3.7 is concrete evidence of market portability: re-pointing the entire pipeline at a different exchange is a configuration change, comprising a new data source and cost model, rather than a change to the agents or the decision layer.

Liquidity. The evaluation trades only large-capitalisation index constituents at the daily close, and the selected configuration is low-turnover, so the resting-order footprint is small at retail size. Price impact, partial fills, and order-book dynamics are not modelled and are discussed as a limitation in Section 6.2.

Execution latency. Because signals are computed once after the close and acted on at the next session, the design tolerates latencies of seconds to minutes and requires no co-location, tick data, or order-book integration, which places it at the opposite end of the infrastructure spectrum from high-frequency trading.

Portfolio scalability and the retail-institutional distinction. The fractional position-sizing rule scales capital mechanically, but the assumption of fills at the closing price without impact bounds the deployable capital relative to average daily volume, so the profile is most defensible at retail-to-moderate assets under management. For a retail investor the system is usable as an end-of-day decision-support overlay on a passive benchmark, without leverage or dedicated infrastructure. For an institutional user the linear scaling and market-agnostic interface make wider universes feasible, but explicit price-impact modelling and the analytical-only leverage caveat of Section 5.2.4 would become binding.

5.4.8. Design implications

Three design implications follow from the experiments. First, the transparency of the decision-agent score, which is a linear combination of agent signals weighted by confidence and by family, is both a theoretical convenience and a practical one. The contribution of each agent on a given day can be read off directly from the score, and the ablation shows that this attribution is consistent with the marginal contribution of that agent to the headline metrics.

Second, the fact that the cost-sensitivity sweep in Table 5.6 preserves a Sharpe ratio strictly above the passive benchmark at every plausible cost configuration suggests that the ensemble is not surviving only thanks to a favourable cost calibration.

Third, the stability of the selected (mode, $c_{\min}$) configuration across the three walk-forward folds indicates that a simple grid-search procedure on a long enough training window is adequate, and that more elaborate parameter-adaptation machinery is not needed at the daily frequency on WIG20.

5.5. Chapter summary

The evaluation reported in this chapter was designed to produce results that survive scrutiny rather than results that look impressive. Parameters were selected on a training window and never revisited, every trade was charged realistic commissions and slippage, and every headline figure was accompanied by a bootstrap confidence interval and a paired significance test.

Under that protocol, the finding is consistent and narrow. The ensemble does not beat passive buy-and-hold on gross total return in either test window, and on the WIG20 panel it trails by eight percentage points. It does, however, lead on every risk-adjusted metric considered, halves the maximum drawdown, reduces the integrated drawdown by nearly two-thirds, and overtakes the

benchmark on total return once the two series are placed on the same ex-ante risk budget. The regime decomposition locates the source of this profile precisely: the ensemble defends during corrections and carries during sideways phases, and pays for both by under-participating in bull legs.

The robustness checks constrain how much of this can be attributed to chance or to arbitrary design choices. The profile survives the removal of any single agent, perturbation and re-estimation of the family weights, every plausible transaction-cost calibration, a change of the selection objective, a three-fold chronological re-split, and a full replication on a second market with a different cost regime. What it does not survive is the demand for conventional statistical significance on a single sub-two-year window, and that limitation is stated plainly rather than glossed over.

One result deserves emphasis because it cuts against the original design intent. The confidence values that the decision agent uses as weights carry no measurable information about signal-level accuracy. The system works, on the risk axis, but not for the reason its architecture suggests: the operative mechanism is the pooling of weakly and negatively correlated signals across methodological families, not precision weighting. Chapter 6 returns to what this implies for the contribution of the thesis and for the work that should follow it.

6. SUMMARY

The aim of this thesis, stated in Section 1.2, was to design a transparent multi-agent decision support system for stock trading on the Warsaw Stock Exchange and to evaluate it under a protocol rigorous enough that its results can be trusted. Both parts of that aim were met, and the second turned out to matter more than the first.

The system itself is deliberately simple: five signal agents, each confined to one methodological family, feed a decision agent that combines their signals into a linear, confidence-weighted score and thresholds it into a trading action. The evaluation around it is not simple. It combines anchored walk-forward tuning, calibrated commissions and slippage, five risk-adjusted metrics, a volatility-matched comparison, bootstrap confidence intervals and paired tests, a decomposition into market regimes, a leave-one-agent-out ablation, sensitivity analyses over the family weights and the selection objective, a cost-sensitivity sweep, an audit of the confidence heuristic, two machine-learning baselines, and a full replication on a second market.

What that evaluation established is narrower than what the project set out to find, and more defensible. The ensemble is not a return enhancer. Passive buy-and-hold earned more on both test windows, because both were dominated by bull regimes and a partially invested strategy cannot keep pace with a fully invested one in such conditions. The ensemble is, however, a risk reducer, and consistently so: it halved the maximum drawdown on WIG20, cut the integrated drawdown by nearly two-thirds, led on every risk-adjusted ratio reported, and overtook the benchmark on total return once both series were placed on the same ex-ante risk budget. Section 3.3 stated three hypotheses; the evidence is consistent with all three on point estimates, and with none of them at conventional levels of statistical significance on a single window of under two years.

6.1. Conclusions

The work supports the following conclusions:

1. the multi-agent system improves risk-adjusted performance rather than gross return, leading buy-and-hold on Sharpe, Sortino, Calmar, Ulcer, and Martin measures while trailing it on cumulative profit in bull-dominated windows,
2. the advantage is concentrated where the design predicts it should be, namely +12.57 percentage points of correction-regime defence and +5.14 percentage points of sideways-regime carry, paid for by underperformance in the bull phase,
3. the profile is not an artefact of any single design choice, since it survives the removal of any one agent, perturbation and re-estimation of the family weights, every plausible cost calibration, a change of selection objective, and a three-fold chronological re-split,
4. the profile is not specific to the Warsaw Stock Exchange, since the FTSE 100 replication reproduces it under a different constituent set, currency, and cost regime that includes an asymmetric buy-side transaction tax,
5. the transparent ensemble remains competitive with an LSTM and a gradient-boosted-trees classifier evaluated under the identical protocol, both of which achieve their apparent edge on the FTSE window by remaining near-fully invested through a bull market,

6. the confidence heuristic does not work as designed, since confidence values carry no measurable information about signal-level accuracy on either market, so the inverse-variance reading of the aggregation rule is a conceptual analogy rather than a calibrated mechanism,
7. the mechanism that does operate is error diversity: pooling weakly and negatively correlated signals across methodological families lowers the variance of the aggregate position irrespective of how the pool is weighted,
8. none of the observed advantages reaches conventional statistical significance on a single sub-two-year test window, and they are therefore reported as economically meaningful rather than as statistically established.

The sixth and seventh conclusions are the ones that were not anticipated at the outset, and they are worth stating plainly. The architecture was motivated by the idea that a confidence value could act as a proxy for the reliability of an agent at a given moment, so that the decision agent would perform something close to precision-weighted averaging. The confidence-validation experiment of Section 5.2.8 shows that this is not what happens. The system works on the risk axis, but for the weaker and more robust of the two reasons that forecast-combination theory offers.

Methodologically, the evaluation pipeline is the more transferable contribution. It closes the four gaps identified in Chapter 2, and it is reusable as a template for similar studies on smaller equity markets, where capital preservation in turbulent periods may matter more than chasing the next bull leg.

6.2. Difficulties and limitations

Several limitations of this study should be kept in mind.

First, although the evaluation spans two markets, both are large-capitalisation blue-chip indices evaluated at daily frequency. Generalisation to smaller and less liquid stocks, to other asset classes, or to intraday frequencies would require a separate evaluation, and the signal agents and the decision agent, which were designed around daily closes, would need to be adapted for intraday use.

Second, although the transaction-cost model captures commissions and slippage and is stress-tested over a wide grid, it does not capture liquidity constraints, partial fills, or the price impact of larger orders. In practice, large orders could move the closing price and would require an order-book-level simulation.

Third, the system does not use fundamental data, news, or alternative data sources, which are known to carry orthogonal information with respect to price-based indicators and would fit naturally into the forecast-combination framework as additional agents with distinct error structures.

Fourth, the decision agent uses fixed family weights and heuristic confidence values. The sensitivity analysis of Section 5.3.2 shows that the headline conclusions are robust to perturbing, reordering, or re-estimating the weights, and that in-sample optimisation does not reliably improve on the fixed prior. Nevertheless, the weights are static over time, and regime-conditional weighting remains untested. Relatedly, Section 5.2.8 shows that the confidence values do not predict signal-level accuracy, so the decision layer cannot claim calibrated precision weighting, and the link to classical forecast-combination theory remains conceptual.

Fifth, the test window, although long enough to cover bull, correction, and sideways regimes, is ultimately a single window of under two years, and the bootstrap tests have limited power to

detect moderate effect sizes on a sample of this size. Point estimates and confidence intervals therefore carry as much of the argument as p -values do.

Sixth, the volatility-matched comparison requires leverage of $\lambda = 2.06$, which is impractical for a typical retail investor on the WSE. That comparison serves as a counterfactual on risk budget rather than as an operational recommendation.

Finally, the study does not include a live-trading exercise. Any claim about realised performance in production would require one.

Beyond these limitations of the study itself, two difficulties shaped the course of the work. The first was computational: the full experiment battery re-runs the backtest hundreds of times across grid points, ablations, weight perturbations, and cost configurations, which was intractable until signal computation was decoupled from portfolio simulation and the agent signals were precomputed once (Algorithm 1). That change reduced the end-to-end runtime by more than an order of magnitude and made the sensitivity analyses affordable.

The second difficulty was interpretive. An early version of the work compared the system against buy-and-hold on cumulative return alone, and on that basis the system looked like a failure. Recognising that the comparison was confounded, because a partially invested strategy was being measured against a fully invested benchmark at roughly twice the volatility, is what motivated the risk-adjusted metric battery, the regime decomposition, and the volatility-matched comparison that now form the core of the evaluation.

6.3. The way forward

Each limitation maps to a direction for further work.

Building on the FTSE 100 validation reported here, the system could be extended to further indices, for example the German DAX, the Italian FTSE MIB, or the US S&P 500, and to smaller-capitalisation or emerging-market segments, in order to map more precisely the market conditions under which the risk-adjusted advantage holds. Intraday frequencies would require a careful revision of both the signal agents and the transaction-cost model, and would expose the system to additional market-microstructure effects [52].

Fundamental and alternative data could be integrated as additional signal agents that share the same base-agent interface, which is straightforward given the modular design described in Chapter 4.

Because static in-sample weight optimisation and inverse-variance estimation add little (Section 5.3.2), the more promising weighting extension is regime-conditional family weights, in which the trend family is upweighted in bull regimes and downweighted in corrections. This is a natural evolution of the current fixed-weight design and would be expected to recover part of the bull-regime shortfall identified in Section 5.2.5.

A complementary direction is to replace the heuristic confidence values with calibrated probabilities, for example by isotonic or Platt-type mappings from indicator magnitudes to empirical hit rates estimated on the training window [49]. Calibrated confidences would let the decision layer implement genuine precision weighting and would upgrade the conceptual link to forecast-combination theory into a testable quantitative one. Given the outcome of Section 5.2.8, this is the single most valuable extension of the present design.

Beyond weighting, the fixed roster of agents could be made adaptive through dynamic agent selection, in which the active pool is chosen per regime or per symbol according to recent forecast accuracy, so that agents contribute only where they have demonstrated skill. More ambitiously, the

linear confidence-weighted score could be replaced by a learning-based ensembling layer, such as a stacked meta-learner or a gating network that learns the combination function from data. The challenge there is to retain the per-agent attribution and interpretability that motivate the present design while admitting a richer, potentially non-linear combiner.

Finally, a walk-forward live-trading exercise on a paper-trading broker interface would close the loop between the backtest and realistic execution, and would test the operational assumptions, namely end-of-day execution at closing prices without market impact, that the backtest takes for granted.

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